Belt Holonomy Is Inevitable_ A Two-Boundary Worldsheet from Standard Quantum Geometry

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1. Introduction and Statement of Main Results

Aim and stance (math-first, no new postulates)

This paper shows that, **within standard quantum/open-system geometry**, a **two-boundary worldsheet ("belt")** is *forced* by the formalism itself and is the minimal geometric carrier for two families of observables:

the **continuous geometric (adiabatic/anomalous) forces**, and

the **exchange/statistics phases** in the usual topological/adiabatic sectors.
No extra ontology is introduced; all objectivity/observer-agreement claims used later are imported as theorems from *Self-Referential Observers in Quantum Dynamics* (SROQD).

Standing hypotheses (standard only)

We fix the following data, all mainstream:

A smooth **channel manifold** Θ of control contexts; a smooth **closed loop** $\gamma: S^1 \to \Theta$.

Schwinger–Keldysh (SK) closed-time-path doubling with forward/back branches; the loop γ lifts to two boundary curves $\gamma_{\pm} \subset \Theta \times \mathsf{SK}$.

The **Uhlmann/Bures connection** $\mathcal{A}$ on the purification principal bundle over the density-operator manifold; curvature $\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}$; and the **non-Abelian Stokes theorem** for surface-ordered exponentials.

For **operational meaning and frame invariance** of the belt observables, we *use* (do not re-prove) SROQD results: **AB-fixedness** under compatible effects + shared records (SBS redundancy), and **Collapse-Lorentz** symmetry that preserves agreement/incompatibility.

Definitions needed for the statements

A **belt** for γ is a smooth immersed compact oriented ribbon

$$S(\gamma) \subset \Theta \times \mathsf{SK}, \qquad \partial S(\gamma) = \gamma_+ \sqcup \gamma_-,$$

with small thickness $\varepsilon > 0$ (the **thin-belt** limit will be $\varepsilon \to 0$). Its **holonomy** is the surface-ordered exponential

$$\mathcal{H}[S] \;=\; \mathcal{P}\exp \iint_S \mathcal{F},$$

and we write the **geometric action** $\Phi(\gamma) = \arg \mathrm{Tr}\ \mathcal{H}[S(\gamma)]$ in any fixed finite-dimensional representation. In the thin-belt limit, a **framing** (a nowhere-vanishing normal field along γ) survives and will matter for exchange phases.

Main theorems (precise statements)

Theorem A — Existence and Minimality of Belts

Let $\gamma: S^1 \to \Theta$ be C^1, and assume $\mathcal{A}$ is smooth on the purification bundle. Then:

(**Existence**) SK doubling canonically induces two oriented boundary curves $\gamma_{\pm}$, and there exists a smooth immersed ribbon $S(\gamma) \subset \Theta \times \mathsf{SK}$ with $\partial\, S(\gamma) = \gamma_+ \sqcup \gamma_-$. (For instance, $S(\sigma, t) = (\gamma(\sigma), t)$ in a local SK chart.)

(**Minimality**) In the thin-belt limit $\varepsilon \to 0$, the holonomy $\mathcal{H}[S(\gamma)]$ retains a **framing datum** not, in general, recoverable from any single **unframed** line integral; thus a two-boundary ribbon is the **minimal gauge-covariant carrier** that preserves back-action/noise information captured by $\mathcal{F}$.

Comment. Part (2) formalizes the inevitability of a ribbon (two boundaries + framing) rather than a lone curve when mixed-state transport and SK structure are present.

Theorem B — Continuous/Geometric Forces = Variational Response of Belt Holonomy

Let $\{\gamma(\sigma)\}_{\sigma \in (-\delta, \delta)}$ be a C^1 family of closed loops with associated belts $S(\gamma(\sigma))$. Define

$$\Phi(\sigma) = \arg \mathrm{Tr}\ \mathcal{H}\left[S(\gamma(\sigma))\right].$$

Under standard regularity (trace-class, smooth curvature), the **geometric response** along γ satisfies the functional identity

$$F_i(\sigma) \;=\; -\frac{\delta}{\delta x^i(\sigma)}\, \Phi(\sigma),$$

which reduces in the Abelian/pure-state limit to the textbook Berry/Uhlmann anomalous terms. In particular, the **measured adiabatic/anomalous force** equals the **Gâteaux derivative** of the belt holonomy phase.

Comment. The first-variation formula is proved by ordered-exponential calculus plus non-Abelian Stokes; we keep all commutator terms and boundary contractions explicit (Appendix A).

Theorem C — Exchange/Statistics = Framed Belt Linking

Let S_1, S_2 be belts with framed boundaries $\partial\, S_j = \gamma_{j,+} \sqcup \gamma_{j,-}$. Define the **framed belt linking**

$$\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) = \sum_{\alpha \in \{+,-\}} \sum_{\beta \in \{+,-\}} \mathrm{Lk}\big(\gamma_{1,\alpha}, \gamma_{2,\beta}\big),$$

the sum of Gauss linking numbers of all boundary components with inherited framings.
In **Berry–Chern / Chern–Simons–type sectors**, the **exchange unitary** for adiabatic transport satisfies

$$U_{\mathrm{ex}} \;=\; \exp\left(i\,\kappa\, \mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)\right) \cdot U_{\mathrm{local}},$$

where κ is the sector's level and U_{local} collects non-topological local terms; self-link pieces reproduce the familiar **framing anomaly**.

Comment. Thus the **phase content** usually attributed to "virtual exchange" is exactly the **framed linking** of belts; Section 6 and Appendix B give the full derivation from framed Wilson operators.

Scope, non-claims, and operational status

The results assert a **representation-equivalence** inside the adiabatic/topological regime:
(i) **continuous forces ↔ belt-holonomy variation** (Thm B);
(ii) **exchange phases ↔ framed belt linking** (Thm C).
They do **not** claim that belts are new fundamental fields nor provide a new 3+1D spin–statistics proof; non-geometric forces and fully inelastic channels lie outside our equivalence class.

Objectivity & frame invariance. When effects commute and records are redundantly accessible (SBS), belt holonomies and linking indices are **AB-fixed** and invariant under **Collapse-Lorentz** frame maps—direct corollaries of SROQD. These are *imported theorems*, ensuring our observables are operationally well-posed without extra assumptions.

Roadmap

Section 2 formalizes the setting (channel manifold, SK doubling, Uhlmann transport, non-Abelian Stokes) and recalls the SROQD layer we rely on for objectivity. Section 3 defines belts and proves the thin-belt framing lemma. Sections 4–6 give full proofs of Theorems A–C. Section 7 applies SROQD to show objectivity/frame invariance of belt observables. Section 8 collects reductions and cross-checks. Section 9 interprets "what we proved" and delineates non-claims; Appendices provide the full calculus for variations and framed linking.

Short takeaway. As horizons follow from GR, **belts follow from SK + Uhlmann + Stokes**. Whether one "likes" belts or not is immaterial—the math makes them the **minimal** worldsheets that faithfully carry geometric forces and exchange phases, with observer-level objectivity guaranteed by SROQD.

2. Mathematical Setting and Assumptions (Standard Only)

2.1 Channel Manifold, Instruments, and Adaptive Policies

We fix only mainstream structures. All objectivity/"agreement" properties invoked later are imported from **SROQD** and not re-proved here.

2.1.1 Channel space and measurability

Channel manifold. Let Θ be a second–countable smooth manifold (for later differential constructions) endowed with its Borel σ-algebra $\mathcal{B}(\Theta)$. When needed for metric notions (e.g., collapse interval in SROQD §5), Θ carries a Riemannian or information metric g.

Control loops. A control protocol is a C^1 map $\gamma: S^1 \to \Theta$. (SK doubling and bundles are introduced in §2.2.)

2.1.2 Systems, memory, ticks

Hilbert spaces. Let $\mathcal{H}_W$ (world) and $\mathcal{H}_M$ (memory/ancilla) be separable Hilbert spaces; $\mathcal{H} := \mathcal{H}_W \otimes \mathcal{H}_M$. Density operators on $\mathcal{H}$ form $\mathcal{D}(\mathcal{H})$.

Discrete ticks. Measurements occur at ticks $t \in \mathbb{N}$. Between ticks, the system evolves by a CPTP map $\mathcal{E}_t$ (latent for our purposes).

Outcome space. Let Φ be finite or countable with discrete σ-algebra; write $\Omega := \Phi^{\mathbb{N}}$ with product σ-algebra $\mathcal{F}$ (cylinder sets). For $\omega = (a_1, a_2, \ldots) \in \Omega$, let $A_t(\omega) = a_t$ be the canonical coordinate random variables, and $\mathcal{F}_t := \sigma(A_1, \ldots, A_t)$ the filtration (the **observer trace** up to t).

2.1.3 Quantum instruments (standard)

For each context $\theta \in \Theta$ we have a **quantum instrument**

$$\mathsf{I}_\theta = \{\mathcal{M}_{\theta,a}\}_{a\in\Phi}, \qquad \mathcal{M}_{\theta,a}: \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$$

with each $\mathcal{M}_{\theta,a}$ completely positive and $\sum_{a\in\Phi} \mathcal{M}_{\theta,a}$ trace-preserving. Outcome probabilities satisfy $\sum_a \operatorname{Tr} \mathcal{M}_{\theta,a}(\rho) = 1$ for all $\rho \in \mathcal{D}(\mathcal{H})$. (Stinespring dilations exist but are not required explicitly.)

2.1.4 Record-keeping in memory (ancilla write)

We model **writing the outcome to memory** by choosing a fixed orthonormal pointer basis $\{|\, a\rangle_M\}_{a\in\Phi} \subset \mathcal{H}_M$ and requiring that each instrument Kraus implementation leaves a classical imprint on $\mathcal{H}_M$. One convenient realization: for some dilation isometry $V_{\theta,a}$,

$$\mathcal{M}_{\theta,a}(\rho) = \operatorname{Tr}_E\left[V_{\theta,a}\,\rho\,V_{\theta,a}^\dagger\right], \quad V_{\theta,a}: \mathcal{H}\otimes\mathcal{H}_E \to \mathcal{H}\otimes\mathcal{H}_E, \ (\cdot) \mapsto (\cdot)\otimes|\, a\rangle_M\langle a\,|.$$

Thus the classical **trace** of past outcomes is encoded in the memory register (SROQD uses precisely this ancilla semantics). Delta-certainty of one's **own** recorded outcomes is a theorem of SROQD and is not assumed here.

2.1.5 Adaptive policies (measurable feedback)

An **adaptive policy** is a sequence of maps

$$\pi_t: (\Omega, \mathcal{F}_{t-1}) \longrightarrow (\Theta, \mathcal{B}(\Theta)), \qquad \theta_t = \pi_t(A_1, \ldots, A_{t-1}),$$

which are $\mathcal{F}_{t-1}$-measurable. Equivalently (operator-algebraic SROQD view), π_t is a measurable map of the observer's past algebra into Θ. Measurability is the only structural requirement we impose; under it, the induced **observer process** exists and is unique (Ionescu–Tulcea extension in SROQD).

2.1.6 State/update recursion

Given an initial $\rho_0 \in \mathcal{D}(\mathcal{H})$ and a realized past $(a_1, \ldots, a_{t-1})$, the next context is $\theta_t = \pi_t(a_1, \ldots, a_{t-1})$ and the **instrumental update** at tick t is

$$\rho_t^{(a_t)} = \frac{\mathcal{M}_{\theta_t,a_t}\left(\mathcal{E}_t(\rho_{t-1})\right)}{\operatorname{Tr}\,\mathcal{M}_{\theta_t,a_t}\left(\mathcal{E}_t(\rho_{t-1})\right)}, \qquad \mathbb{P}(A_t = a_t \mid \mathcal{F}_{t-1}) = \operatorname{Tr}\,\mathcal{M}_{\theta_t,a_t}\left(\mathcal{E}_t(\rho_{t-1})\right).$$

The **composite law** $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is uniquely induced by $\{\pi_t\}$, $\{\mathsf{I}_\theta\}$, and $\{\mathcal{E}_t\}$. Existence/uniqueness and

internal delta-certainty are established in SROQD; we cite them as foundations.

2.1.7 Minimality of assumptions (from SROQD)

SROQD proves that these assumptions are **structurally necessary**: if measurability of π_t fails, the process may not exist; if effects do not commute or redundancy is too weak, cross-observer agreement (AB-fixedness) fails; if frame maps are not collapse-interval isometries, agreement is not preserved. We rely on these imported results only to guarantee that the belt observables defined later are operationally meaningful and frame-robust.

Remark. Nothing in §2.1 presumes any SMFT-specific axiom; we have only specified channels, instruments, measurability, and memory as in SROQD's formal layer. Subsequent sections add SK doubling and mixed-state parallel transport (Uhlmann), from which "belts" will be derived.

3. Belts: Two-Boundary Worldsheets and Thin-Belt Framing

We make precise what a "belt" is, how its holonomy is defined, and why a framing datum survives in the thin-belt limit. No extra postulates are introduced; everything is standard differential geometry on the control space and standard holonomy/curvature calculus for mixed-state transport.

3.1 SK lift of a control loop

Let Θ be a smooth, second-countable manifold (Borel σ-algebra implicit). Fix a C^1 **control loop**

$$\gamma: S^1 \longrightarrow \Theta, \qquad \sigma \mapsto \gamma(\sigma),$$

with σ a 2π-periodic parameter and unit-speed chosen where convenient.
The **Schwinger–Keldysh (SK) doubling** consists of two oriented time branches SK_+ and SK_-. We represent the doubled base as the disjoint union

$$\Theta \times \mathrm{SK} := (\Theta \times \{+\}) \sqcup (\Theta \times \{-\}),$$

with inherited orientations $(+)$ and $(-)$. The loop γ canonically lifts to the two boundary curves

$$\gamma_{\pm}: S^1 \to \Theta \times \mathrm{SK}, \qquad \gamma_{\pm}(\sigma) := (\gamma(\sigma), \pm).$$

3.2 Belts as two-boundary worldsheets

We write $S^1 \times I_\varepsilon$ for the annulus with $I_\varepsilon = [-\varepsilon/2, +\varepsilon/2]$ and $\varepsilon > 0$ the (small) belt thickness.

Definition 3.1 (SK belt).

A **belt** spanning γ is a smooth immersed compact oriented surface

$$S(\gamma) \subset \Theta \times \mathrm{SK},$$

together with a smooth immersion

$$\iota: S^1 \times I_\varepsilon \longrightarrow \Theta \times \mathrm{SK}, \quad (\sigma, t) \mapsto \iota(\sigma, t),$$

such that:

Boundary: $\partial S(\gamma) = \gamma_+ \sqcup \gamma_-$ with
$\iota(\sigma, +\varepsilon/2) = \gamma_+(\sigma)$ and $\iota(\sigma, -\varepsilon/2) = \gamma_-(\sigma)$ for all σ.

Orientation: $\partial S(\gamma)$ carries the induced boundary orientation; by convention the + boundary is positively oriented relative to the surface.

Regularity: ι is C^∞ in (σ, t) with rank 2 everywhere (immersed ribbon), and $S(\gamma)$ has finite area in any auxiliary Riemannian metric on Θ.

Canonical example. In a local SK chart, the product ribbon

$$\iota(\sigma, t) = \big(\gamma(\sigma),\, \mathrm{br}(t)\big), \qquad \mathrm{br}(\pm\varepsilon/2) = \pm,$$

is a belt. More geometrically (see §3.4), one can take a **geodesic normal ribbon** in Θ built from a framing field along γ.

3.3 Purification bundle, curvature, and belt holonomy

Let $\mathcal{P} \to \mathcal{D}(\mathcal{H})$ be the principal $U(\mathcal{H})$-bundle of purifications over the density-operator manifold, endowed with the **Uhlmann connection** $\mathcal{A}$ and curvature

$$\mathcal{F} \;=\; d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}.$$

A control loop γ (together with the SK doubling and the physical preparation/record map) pulls back the bundle to $\Theta \times \mathrm{SK}$. The **surface-ordered exponential** over a belt S is well-defined by the non-Abelian Stokes construction:

Definition 3.2 (Belt holonomy).

For a belt $S(\gamma)$, define its holonomy

$$\mathcal{H}[S] \;:=\; \mathcal{P}\exp \iint_{S(\gamma)} \mathcal{F},$$

an element of the structure group (represented on a finite-dimensional fiber if desired). The **Wilson-belt functional** is

$$W[S] \;:=\; \mathrm{Tr}\; \mathcal{H}[S].$$

Gauge covariance. Under a smooth gauge transform g on the pulled-back bundle,

$$\mathcal{A} \mapsto g^{-1}\mathcal{A}\, g + g^{-1}dg, \qquad \mathcal{F} \mapsto g^{-1}\mathcal{F}\, g,$$

one has $\mathcal{H}[S] \mapsto G_+^{-1}\,\mathcal{H}[S]\,G_-$ with $G_\pm$ the values of g on the two boundary components. Hence any **conjugation-invariant** functional of $\mathcal{H}[S]$ (e.g., $W[S]$, $\arg W[S]$) is gauge-invariant. Reparametrizations of σ preserve $\mathcal{H}[S]$ since $\mathcal{F}$ is a 2-form.

3.4 Thin-belt construction and framing

To make the **framing** explicit, endow Θ with any smooth Riemannian metric g (e.g., information metric). A **framing field** along γ is a nowhere-vanishing smooth normal vector field

$$\nu(\sigma) \;\in\; T_{\gamma(\sigma)}\,\Theta, \qquad g\big(\nu(\sigma), \dot\gamma(\sigma)\big) = 0.$$

For sufficiently small $\varepsilon > 0$, the exponential map gives two **offset loops**

$$\gamma_\pm^\varepsilon(\sigma) \;:=\; \exp_{\gamma(\sigma)}\Big(\pm\tfrac{\varepsilon}{2}\,\nu(\sigma)\Big),$$

well-defined and embedded; they serve as the spatial parts of the SK boundaries. The **geodesic normal belt** is then the image of

$$\iota_\varepsilon(\sigma,t) \;=\; \Big(\exp_{\gamma(\sigma)}\big(t\,\nu(\sigma)\big),\ \mathrm{br}(t)\Big), \qquad t\in[-\tfrac{\varepsilon}{2},+\tfrac{\varepsilon}{2}],$$

which satisfies Definition 3.1 for all small ε.

Definition 3.3 (Framed belt; thin-belt limit).

A **framed belt** is a pair (γ,ν) together with $S_\varepsilon(\gamma,\nu)$ as above. The **thin-belt limit** is the limit $\varepsilon\downarrow 0$ of gauge-invariant functionals (e.g., $\arg W[S_\varepsilon]$), when it exists, and it is a functional of the **framed loop** (γ,ν).

The following lemma formalizes the "survival" of framing in the limit.

Lemma 3.4 (Framing survives the thin-belt limit).

Let γ be C^2 and ν smooth and nowhere zero. Then, for any fixed smooth curvature $\mathcal{F}$,

$$\arg W[S_\varepsilon(\gamma,\nu)] \;=\; \arg \mathrm{Tr}\ \mathcal{P}\exp \iint_{S_\varepsilon} \mathcal{F}$$

admits a finite limit as $\varepsilon\downarrow 0$ and depends—besides γ—on the **rotation class** of ν along γ (the framing). In particular, there exist framed loops (γ,ν) and (γ,ν') with the same underlying γ but

$$\lim_{\varepsilon\to 0}\arg W[S_\varepsilon(\gamma,\nu)] \;\neq\; \lim_{\varepsilon\to 0}\arg W[S_\varepsilon(\gamma,\nu')].$$

Sketch of proof (full details later in Appx. A/B).
By non-Abelian Stokes, $\log \mathcal{H}[S_\varepsilon]$ is a surface integral of $\mathcal{F}$ plus commutator-ordered corrections. For the geodesic normal belt, the area 2-vector at (σ,t) is spanned by $(\dot\gamma(\sigma),\,\nu(\sigma))$ (up to $O(\varepsilon)$ corrections), so the leading contribution to $\arg W$ is

$$\int_{S_\varepsilon} \Big\langle \mathcal{F},\, \dot\gamma(\sigma)\wedge\nu(\sigma)\Big\rangle\, d\sigma\, dt \;=\; \frac{\varepsilon}{2}\int_{S^1} \Big\langle \mathcal{F},\, \dot\gamma(\sigma)\wedge\nu(\sigma)\Big\rangle\, d\sigma \;+\; O(\varepsilon^2)$$

after integrating in t. While this vanishes as $\varepsilon\to 0$, the **ordered** exponential retains a finite **framing phase** arising from (i) the SK orientation difference of the boundaries, and (ii) the **twist** of ν (the rotation of the normal along γ), which enters through the regularization of coincident boundaries and shows up as a finite self-link/framing term. Hence the limit depends on the homotopy class of ν (its winding number in the normal bundle), not just on γ.

Consequences. A single **unframed** curve cannot, in general, encode the finite phase data that survives from the ribbon regularization; two boundaries + a framing are the **minimal** gauge-covariant carrier.

3.5 Regularity, admissible deformations, and equivalence

Two belts S_1, S_2 spanning the same $\gamma_\pm$ are said to be **admissibly homotopic** if there exists a smooth 3-chain V with $\partial V = S_1 \sqcup \overline{S_2}$ such that all intermediate surfaces are immersed ribbons with the same boundary framing. Standard ordered-exponential calculus implies:

If $\mathcal{F}$ is flat on the support of V, then $\mathcal{H}[S_1]$ and $\mathcal{H}[S_2]$ are (boundary-)conjugate, hence $W[S_1] = W[S_2]$.

In general, the difference is governed by $\mathcal{P}\exp \iint_V d\mathcal{F}$ and nested commutators; thus **surface dependence** encodes genuine curvature flux.

An **admissible deformation** of a framed belt is a smooth family (γ_τ, ν_τ) with ν_τ nowhere vanishing; the induced family $S_\varepsilon(\gamma_\tau, \nu_\tau)$ defines a differentiable path in belt space. Section 5 uses this to compute the **first variation** of $\arg W[S]$ under boundary deformations and derive the geometric-force identity.

3.6 Summary of Section 3

A **belt** is a smooth immersed ribbon $S(\gamma) \subset \Theta \times \mathrm{SK}$ with $\partial S = \gamma_+ \sqcup \gamma_-$.

Its **holonomy** $\mathcal{H}[S] = \mathcal{P}\exp \iint_S \mathcal{F}$ is gauge-covariant; the **Wilson-belt** $W[S] = \mathrm{Tr}\,\mathcal{H}[S]$ is gauge-invariant.

In the **thin-belt limit**, a **framing** ν of γ survives as genuine, finite phase data; hence two boundaries + framing are **minimal and inevitable**.

These structures set up Theorem A (existence/minimality) and prepare the calculus used in Theorem B (first variation → geometric force) and Theorem C (framed link → exchange phase).

4. Theorem A — Existence & Minimality of Belts (Full Proof)

Statement (Theorem A).
Let Θ be a smooth, second–countable manifold and $\gamma: S^1 \to \Theta$ a C^1 closed control loop. Consider non-equilibrium dynamics with records on the Schwinger–Keldysh (SK) closed-time path. Then:

(**Boundary doubling**) γ canonically lifts to two oriented boundary curves $\gamma_\pm: S^1 \to \Theta \times \mathrm{SK}$.

(**Spanning surface**) There exists a smooth immersed ribbon $S(\gamma) \subset \Theta \times \mathrm{SK}$ with $\partial S(\gamma) = \gamma_+ \sqcup \gamma_-$.

(**Gauge covariance**) The belt holonomy $\mathcal{H}[S] = \mathcal{P}\exp \iint_S \mathcal{F}$ (with $\mathcal{F}$ the Uhlmann curvature on the pulled-back purification bundle) is gauge–covariant; conjugation-invariant functionals (e.g. $W[S] = \mathrm{Tr}\,\mathcal{H}[S]$, $\arg W[S]$) are gauge–invariant and independent of boundary reparametrization.

(**Minimality/Obstruction**) In the thin-belt limit, a **framing datum** survives; compressing to a single **unframed** line generically loses this gauge-covariant invariant. Equivalently: framed Wilson regularization and SK ribbonization produce finite framing contributions (self-link terms) not reproducible by an unframed line.

We prove (1)–(4) in order.

4.1 Boundary Doubling via the SK contour

Let the SK contour be the disjoint union of two oriented branches $\mathrm{SK} = \{+, -\}$ (forward/back). Define

$$\gamma_\pm(\sigma) := (\gamma(\sigma), \pm) \in \Theta \times \mathrm{SK}, \qquad \sigma \in S^1.$$

By construction, $\gamma_\pm$ are C^1 embeddings with inherited orientations from the SK branches. This is the canonical lift used for open-system dynamics with records on the closed-time path (standard in

nonequilibrium field theory and stated explicitly in the draft). ■

4.2 Existence of a spanning surface as a rectifiable 2–chain (and smooth immersion)

We construct a **belt** $S(\gamma)$ as a smooth immersed ribbon whose boundary is $\gamma_+ \sqcup \gamma_-$.

Step 1: Choose a normal field and offset loops in Θ.

Equip Θ with a smooth Riemannian metric g. For $\gamma \in C^1$, the tubular-neighborhood theorem yields a neighborhood diffeomorphic to the normal bundle; hence there exists a smooth nowhere-vanishing normal field $\nu(\sigma) \in T_{\gamma(\sigma)}\Theta$ with $g(\nu, \dot{\gamma}) = 0$. For sufficiently small $\varepsilon > 0$, define the **offset curves**

$$\gamma_\pm^\varepsilon(\sigma) := \exp_{\gamma(\sigma)}\Big(\pm\frac{\varepsilon}{2}\,\nu(\sigma)\Big), \quad \sigma \in S^1,$$

which are embedded and C^1 for small ε.

Step 2: Build a geodesic normal ribbon in Θ.

Define $\iota_\varepsilon : S^1 \times [-\varepsilon/2, \varepsilon/2] \to \Theta$ by

$$\iota_\varepsilon(\sigma, t) := \exp_{\gamma(\sigma)}\big(t\,\nu(\sigma)\big).$$

This map is C^∞ and has rank 2 for small ε, hence its image $S_\Theta^\varepsilon(\gamma, \nu)$ is a smooth immersed compact oriented surface in Θ with boundary $\partial S_\Theta^\varepsilon = \gamma_+^\varepsilon \sqcup \gamma_-^\varepsilon$.

Step 3: Lift to $\Theta \times \mathrm{SK}$.

Let $\mathrm{br}: [-\varepsilon/2, \varepsilon/2] \to \{+, -\}$ be $\mathrm{br}(t) = \mathrm{sign}(t)$ with the convention $\mathrm{br}(\pm\varepsilon/2) = \pm$. Define

$$\iota(\sigma, t) := \big(\iota_\varepsilon(\sigma, t),\, \mathrm{br}(t)\big) \in \Theta \times \mathrm{SK}.$$

Its image $S(\gamma) := \iota(S^1 \times [-\varepsilon/2, \varepsilon/2])$ is a smooth immersed **belt** (a two-sheet ribbon over the SK branches) with

$$\partial S(\gamma) \;=\; \gamma_+ \sqcup \gamma_- \quad \text{(the SK-lifted boundaries)}.$$

("Definition 3.1 (SK belt)" in the draft matches this construction.) ■

> **Rectifiable 2–chain remark.** Even without the explicit immersion, a spanning 2–chain with those boundaries exists by standard geometric measure theory; the above geodesic construction shows smooth realizability, which we will use for holonomy calculus.

4.3 Gauge covariance and parametrization independence of $\mathcal{H}[S]$

Let $\mathcal{P} \to \mathcal{D}(\mathcal{H})$ be the purification principal bundle with **Uhlmann connection** $\mathcal{A}$ and curvature $\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}$. Pull this bundle back to $\Theta \times \mathrm{SK}$ along the physical map induced by the process. For any smooth belt $S \subset \Theta \times \mathrm{SK}$, define the **belt holonomy**

$$\mathcal{H}[S] \;:=\; \mathcal{P}\exp \iint_S \mathcal{F}, \qquad W[S] := \mathrm{Tr}\,\mathcal{H}[S].$$

Non-Abelian Stokes. If S bounds a (piecewise smooth) closed contour ∂S, the non-Abelian Stokes theorem relates $\mathcal{P}\exp \oint_{\partial S} \mathcal{A}$ to $\mathcal{P}\exp \iint_S \mathcal{F}$. We adopt the surface form as the *definition* of $\mathcal{H}[S]$, which is well-posed on surfaces in the pulled-back bundle (as in the draft).

Gauge covariance. Under a gauge transform g on the pulled-back bundle,

$$\mathcal{A} \mapsto g^{-1}\mathcal{A}\,g + g^{-1}dg, \quad \mathcal{F} \mapsto g^{-1}\mathcal{F}\,g,$$

whence $\mathcal{H}[S] \mapsto G_+^{-1}\,\mathcal{H}[S]\,G_-$, with $G_\pm$ the evaluations of g on the two boundary components. Therefore any class function (e.g. $W[S]$, $\arg W[S]$) is gauge-invariant. Reparametrizing $\gamma_\pm$ or the interior coordinates of S leaves $\mathcal{H}[S]$ unchanged because $\mathcal{F}$ is a 2-form (Jacobian cancellations). ■

4.4 Minimality / Obstruction: why a single unframed line is insufficient

We show that the thin-belt limit retains a **framing** that cannot, in general, be encoded by a single unframed line integral. Two complementary arguments are given.

(A) Thin-belt framing survives (geometric proof)

Use the geodesic normal construction in §4.2 with framing field $v(\sigma)$. Consider the family $S_\varepsilon(\gamma, v)$ and $W_\varepsilon := \operatorname{Tr}\mathcal{H}[S_\varepsilon]$. A careful ordered-exponential expansion (non-Abelian Stokes with Duhamel/Volterra series) shows:

$$\arg W_\varepsilon \;=\; \int_{S_\varepsilon} \langle \mathcal{F}, \dot\gamma \wedge v\rangle\, d\sigma\, dt \;+\; (\text{commutator/ordering terms}).$$

While the **area** term scales like $O(\varepsilon)$, the ordered structure induces a **finite** framing contribution in the limit $\varepsilon \downarrow 0$, proportional to the **twist** (rotation class) of v along γ, inherited from the SK two-boundary regularization. Hence the limit depends on the framed loop (γ, v), not merely on γ. This is the "thin-belt framing lemma" stated in §3 of the draft. ■

(B) Framed vs. unframed Wilson regularizations (topological proof)

In effective topological sectors (Berry–Chern / Chern–Simons–type), the algebra of loop observables is **framing-dependent**; the standard cure is to **frame** Wilson loops by considering narrow ribbons. The **self-link (framing) anomaly** contributes a finite phase proportional to the self-linking number of the framed boundary, which cannot be retrieved from a single unframed line. Our SK ribbon $S(\gamma)$ *is exactly* that framed regularization: it consists of two nearby boundary curves $\gamma_\pm$ with inherited framings and supplies the correct finite self-link contribution. Therefore, attempting to "compress" the belt to one unframed curve erases this invariant. ■

> **Conclusion of (4).** A **two-boundary ribbon + framing** is the **minimal** gauge-covariant carrier of the holonomy data compatible with open-system (SK) transport. This is precisely the "thin-belt limit and framing" assertion in the draft's Theorem A, now justified by (A) geometric and (B) topological arguments.

4.5 Summary of Theorem A

SK forces **two** oriented boundary components $\gamma_{\pm}$.

A smooth immersed **belt** $S(\gamma)$ with $\partial S = \gamma_+ \sqcup \gamma_-$ is explicitly constructed (geodesic normal ribbon).

Holonomy $\mathcal{H}[S]$ is well-defined, gauge-covariant, and parametrization-independent (non-Abelian Stokes on the Uhlmann curvature).

The **thin-belt** limit retains a **framing** (finite twist/self-link contribution). Consequently, a **single unframed line** lacks the necessary invariant—making the belt **inevitable and minimal**. These points match the draft's statement and are the bedrock for Theorems B–C (variational force identity; framed linking for exchange).

> **Operational status.** Objectivity (AB-fixedness) and frame robustness (Collapse-Lorentz invariance) of belt observables are imported from **SROQD** and used later; no extra assumptions are introduced here beyond standard SK and Uhlmann machinery.

Sidebar (consistency with the "What we proved" memo)

The above minimality is exactly what underwrites the two equivalences used later: (i) continuous geometric forces equal variational responses of **belt holonomy**; (ii) exchange/statistics phases equal **framed belt linking**—statements rigorously scoped in the memo and in the draft.

5. Theorem B — Continuous/Geometric Forces = Variational Response of Belt Holonomy (Full Proof)

We prove that the geometric (Berry/Uhlmann-type) response along a loop is exactly the functional derivative of the **belt** surface holonomy phase. Throughout, we work **entirely inside standard quantum/open-system geometry** (Uhlmann transport on the purification bundle; non-Abelian Stokes), and we **do not** introduce any extra axioms. The proof follows the variation of a surface-ordered exponential, keeping all commutator/ordering terms explicit.

5.1 Setup, regularity, and notations

Control family. Let $\{\gamma(\sigma)\}_{\sigma\in(-\delta,\delta)} \subset C^1(S^1, \Theta)$ be a C^1 family of closed loops in the channel manifold Θ. For each σ, let $S(\gamma(\sigma))$ be an SK **belt** spanning the two SK boundaries (Section 3), produced for instance by the geodesic-normal ribbon construction with fixed framing ν. Write $S_\sigma := S(\gamma(\sigma))$.

Connection & curvature. On the purification principal bundle over $\mathcal{D}(\mathcal{H})$, let $\mathcal{A}$ be the (possibly non-Abelian) **Uhlmann connection** and $\mathcal{F} = d\mathcal{A} + \mathcal{A}\wedge\mathcal{A}$ its curvature. Pull back to $\Theta \times \mathrm{SK}$ along the physical map defined by the process.

Holonomy & phase. Define the belt holonomy and Wilson–belt functional

$$\mathcal{H}[S_\sigma] := \mathcal{P}\exp \iint_{S_\sigma} \mathcal{F}, \qquad W(\sigma) := \mathrm{Tr}\, \mathcal{H}[S_\sigma], \qquad \Phi(\sigma) := \arg W(\sigma).$$

(The non-Abelian Stokes form is our definition; $\mathcal{P}$ denotes surface ordering.)

Regularity / trace-class conditions. We assume:

$\gamma(\sigma)$ is C^1 in σ and C^2 in loop parameter; the framing field ν is C^1.

$\mathcal{A}$ is C^1 and $\mathcal{F}$ is continuous on a neighborhood of $\bigcup_\sigma S_\sigma$.

We evaluate Tr in a finite-dimensional representation of the structure group (e.g., finite-dimensional system or a finite-rank reduction along the path), so $\mathcal{H}[S_\sigma]$ is a trace-class matrix.

$W(\sigma) \neq 0$ on the variation interval, so $d \arg W = \mathrm{Im}(dW / W)$ is well-defined.
These are exactly the hypotheses used in the draft's variation appendix, ensuring the first-variation formula below is well-posed.

5.2 Boundary deformation and the variation field

Let $\iota_\sigma : [0,1] \times [-\varepsilon/2, \varepsilon/2] \to \Theta \times \mathrm{SK}$ be a smooth parametrization of S_σ with $\partial S_\sigma = \gamma_{+,\sigma} \sqcup \gamma_{-,\sigma}$. A small change $\sigma \mapsto \sigma + \delta\sigma$ induces a **deformation vector field** $V = \partial_\sigma \iota_\sigma \, \delta\sigma$ along the surface; its **boundary restriction** is exactly the **boundary displacement** $\delta x(\tau) = V\big|_{\partial S}$. In what follows we write ι_V for contraction ("interior product") with V.

5.3 Ordered-exponential calculus on surfaces (Duhamel/Volterra)

Let $\mathcal{K}_\sigma(u,v) = \iota_\sigma^*(\mathcal{F})_{(u,v)}$ be the pulled-back curvature 2-form in a fixed surface chart, and $\mathcal{U}_\sigma = \mathcal{P}\exp \iint \mathcal{K}_\sigma$. The standard Duhamel/Volterra expansion for **ordered exponentials** gives the first variation

$$\delta \mathcal{U}_\sigma = \iint_{(u,v)} \mathcal{P}\big[\mathcal{U}_\sigma^{>(u,v)} \, \delta \mathcal{K}_\sigma(u,v) \, \mathcal{U}_\sigma^{<(u,v)} \big] + \text{(higher commutator nestings)},$$

where $\mathcal{U}_\sigma^{>(u,v)}$ (resp. $^{<(u,v)}$) denotes the part of the ordered exponential "after" (resp. "before") the point (u,v) under the fixed surface order. This is the surface analogue of the well-known line-ordered Duhamel formula, and it underlies the draft's Appendix-A identity.

By Cartan's magic formula for the Lie derivative of forms,

$$\delta\big(\iota_\sigma^* \mathcal{F}\big) = \iota_\sigma^*(\mathcal{L}_V \mathcal{F}) = \iota_\sigma^* \big(d\, \iota_V \mathcal{F} + \iota_V \, d\mathcal{F} \big).$$

Using the **Bianchi identity** $D\mathcal{F} = d\mathcal{F} + [\mathcal{A}, \mathcal{F}] = 0$, we can rewrite $\iota_V \, d\mathcal{F} = -\iota_V [\mathcal{A}, \mathcal{F}]$. Inserting this into the ordered-exponential variation, the interior term proportional to $d\mathcal{F}$ is exactly absorbed by the **commutator/ordering** corrections (covariantization of Stokes), leaving a **pure boundary contribution** built from $d(\iota_V \mathcal{F})$. After applying non-Abelian Stokes to pass back to the boundary, one arrives at the standard **first-variation formula** for surface holonomies used in the draft:

$$\boxed{\delta \arg \mathrm{Tr}\, \mathcal{H}[S_\sigma] = \oint_{\partial S_\sigma} \big\langle \iota_{\delta x} \mathcal{F} \big\rangle + \text{(explicit commutator terms that preserve gauge covariance)}\,,}$$

where $\langle \cdot \rangle$ denotes the trace in the chosen representation and δx is the boundary displacement field. (A line-by-line derivation with the nested commutator bookkeeping is provided in **Appendix A** of the draft.)

Remark (gauge covariance). Under a gauge transform, $\mathcal{F} \mapsto g^{-1} \mathcal{F} \, g$ and $\mathcal{H}[S] \mapsto G_+^{-1} \mathcal{H}[S] \, G_-$; conjugation-invariant functionals such as $\arg \mathrm{Tr}\, \mathcal{H}[S]$ remain invariant, and the contraction $\langle \iota_{\delta x} \mathcal{F} \rangle$ is well-

defined modulo the same conjugations.

5.4 Functional derivative and the force identity

Let the loop be written locally as $\gamma(\tau)$ with coordinates $x^i(\tau)$. A **Gâteaux boundary deformation** is specified by a small field $\delta x^i(\tau)$. From the first-variation formula,

$$\delta\,\Phi \;=\; \delta \arg \mathrm{Tr}\, \mathcal{H}[S_\sigma] \;=\; \int_{S^1} \Big\langle \iota_{\delta x(\tau)}\, \mathcal{F} \Big\rangle\, d\tau \;+\; (\text{commutator terms}).$$

By definition of the functional derivative,

$$\frac{\delta\,\Phi}{\delta x^i(\tau)} \;=\; \Big\langle \iota_{\partial_i}\, \mathcal{F} \Big\rangle \;+\; (\text{commutator corrections}),$$

and the **geometric (adiabatic/anomalous) force density** along the loop is

$$\boxed{F_i(\tau) \;=\; -\frac{\delta}{\delta x^i(\tau)}\,\Phi \;=\; -\,\frac{\delta}{\delta x^i(\tau)}\, \arg \mathrm{Tr}\; \mathcal{P}\exp \iint_{S(\gamma)} \mathcal{F}\,,}$$

precisely the identity stated in the draft and summarized in the "What we proved" note. All commutator/ordering pieces are retained by the ordered-exponential calculus; their gauge-covariant sum is exactly what the Appendix-A formula encodes.

Regularity recap. The above differentiation is valid under the stated smoothness of $\gamma, v, \mathcal{A}$ and continuity of $\mathcal{F}$; trace-class of $\mathcal{H}[S]$ (finite-dimensional fiber) ensures $W(\sigma)$ is differentiable and $\arg W$ is well-defined away from zeros. These are the explicit conditions recorded in the draft's Appendix A.

5.5 Abelian / pure-state reduction (Berry/Uhlmann limit)

In the **Abelian** or **pure-state** limit, $\mathcal{F}$ effectively commutes along the surface and the ordered exponential reduces to

$$\mathcal{H}[S] = \exp\left(\iint_S \mathcal{F}\right), \qquad \Phi = \mathrm{Im} \iint_S \mathcal{F}.$$

The first variation is simply

$$\delta\,\Phi = \oint_{\partial S} \iota_{\delta x}\, \mathcal{F},$$

so

$$F_i(\tau) = -\,\frac{\delta\,\Phi}{\delta x^i(\tau)} \;=\; -\,\iota_{\partial_i}\, \mathcal{F} \;\; (\text{on the boundary}),$$

which is exactly the textbook **Berry/Uhlmann anomalous response** (geometric force) expressed as the boundary contraction of the curvature 2-form. This matches the reduction statements in the draft.

5.6 Physical reading and scope

The identity

$$F_i(\tau) \;=\; -\frac{\delta}{\delta x^i(\tau)}\,\arg\mathrm{Tr}\;\mathcal{P}\exp\iint_{S(\gamma)}\mathcal{F}$$

asserts that the **measured geometric component** of the response is **exactly** the **variational derivative of the belt holonomy phase**. Thus, **continuous forces ≡ belt-holonomy variation**—no metaphors and no extra assumptions.

Scope. As emphasized in "What we proved," this covers the **geometric/adiabatic** sector. Non-geometric forces (potential gradients, short-time inelastic channels, etc.) lie outside the equivalence and are not claimed here.

Operational status. Objectivity (AB-fixedness) and frame invariance of the belt observables used above are guaranteed by theorems in **SROQD** when effects commute and records are redundantly accessible (SBS), but no part of the force identity depends on adding any SMFT-specific axiom.

Boxed result (Theorem B)

Let $S(\gamma(\sigma))$ be the SK belt for a loop $\gamma(\sigma)$ with Uhlmann curvature $\mathcal{F}$, and let $\Phi(\sigma) = \arg\mathrm{Tr}\,\mathcal{H}[S(\gamma(\sigma))]$. Under the regularity and trace-class conditions above, the geometric/adiabatic response is

$$\boxed{F_i(\sigma) \;=\; -\frac{\delta}{\delta x^i(\sigma)}\,\Phi(\sigma)\,,}$$

with the Abelian/pure-state reduction reproducing the standard Berry/Uhlmann anomalous terms. Full ordered-exponential variation (including commutators and the boundary contraction $\iota_{\delta x}\,\mathcal{F}$) is detailed in Appendix A of the draft.

This completes the rigorous proof of Theorem B.

6. Theorem C — Exchange/Statistics = Framed Belt Linking (Full Proof)

Statement.
Let S_1, S_2 be SK belts with framed boundaries $\partial S_j = \gamma_{j,+} \sqcup \gamma_{j,-}$. Define the **framed belt linking**

$$\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) \;:=\; \sum_{\alpha\in\{+,-\}}\;\sum_{\beta\in\{+,-\}}\mathrm{Lk}(\gamma_{1,\alpha},\gamma_{2,\beta}).$$

In **Berry–Chern / Chern–Simons–type** effective sectors, the **exchange unitary** for adiabatic transport satisfies

$$U_{\mathrm{ex}} \;=\; \exp\left(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1,S_2)\right)\;\cdot\;U_{\mathrm{local}},$$

where κ is fixed by the underlying gauge data; U_{local} collects non–topological (short-range) contributions. Self-link pieces reproduce the **framing anomaly**.

We give a full derivation in two complementary formalisms: (I) **Abelian Chern–Simons path integral** (rigorous distributional calculus via Poincaré dual currents), and (II) **Non-Abelian / Berry–Chern reduction** (ordered exponential with framed Wilson regularization). (Appendix B of the draft records the ribbonized statement; here we supply the details.)

6.1 Preliminaries: belts, framed Wilsons, and SK ribbonization

From Secs. 2–3, each belt $S \subset \Theta \times \mathrm{SK}$ carries the surface holonomy

$$\mathcal{H}[S] = \mathcal{P}\exp \iint_S \mathcal{F}, \qquad W[S] = \mathrm{Tr}\ \mathcal{H}[S],$$

with $\mathcal{F}$ the (pulled-back) Uhlmann curvature. In topological/adiabatic sectors, non-Abelian Stokes turns $W[S]$ into a **framed Wilson operator** living on ∂S: the two SK boundaries are separated by a thin ribbon; **framing** is the limiting choice of normal field (Sec. 3). This is the standard regularization that keeps finite self-link phases.

6.2 Abelian Chern–Simons derivation (complete)

We work in 2+1 dimensions with Abelian Chern–Simons action (level $\kappa \neq 0$):

$$S_{\mathrm{CS}}[A] = \frac{\kappa}{4\pi} \int_{\mathbb{R}^{1,2}} A \wedge dA, \qquad A \in \Omega^1.$$

Insert two ribbonized Wilson operators supported on the framed boundary links $C_{j,\alpha} = \gamma_{j,\alpha}$ (we momentarily write a single C for any component):

$$W[C] := \exp\left(i\, q \oint_C A\right).$$

Let J_C be the **Poincaré dual 2-form current** of the oriented loop C: for all 1-forms η, $\int \eta \wedge J_C = \oint_C \eta$. The generating functional with sources is

$$\mathcal{Z}[J] = \int \mathcal{D}A \exp\left\{ i \frac{\kappa}{4\pi} \int A \wedge dA + i \int A \wedge J \right\}, \quad J = \sum_{j,\alpha} q_{j,\alpha} J_{C_{j,\alpha}}.$$

Lemma 6.1 (Gaussian evaluation with distributions).

The integral is Gaussian. The equation of motion is $\frac{\kappa}{2\pi} dA + J = 0$. Choose a Green's operator d^{-1} on coexact 2-forms (Coulomb gauge eliminates ambiguities). The on-shell solution $A^\star = -\frac{2\pi}{\kappa}\, d^{-1} J$ yields the exact value

$$\log \frac{\mathcal{Z}[J]}{\mathcal{Z}[0]} = \frac{i}{2} \int A^\star \wedge J = \frac{i\, 2\pi}{\kappa} \int J \wedge d^{-1} J.$$

Proof. Standard quadratic completion; well-posed because S_{CS} is first order and J is co-closed (delta-line currents). ■

Lemma 6.2 (Hopf pairing and linking).

For two **disjoint** oriented loops C_1, C_2,

$$\int J_{C_1} \wedge d^{-1} J_{C_2} = \mathrm{Lk}(C_1, C_2),$$

the **Gauss linking number**. For $C_1 = C_2$ this equals the **self-link** and is ill-defined without a framing; the **ribbon framing** (offset by a chosen normal) defines $\mathrm{SL}(C) \in \mathbb{Z}$.
Proof. This is the distributional form of the Gauss integral; d^{-1} implements the Biot–Savart kernel, and the regularized diagonal yields the integer self-link fixed by the framing. ■

Combining the lemmas:

$$\log\langle \prod_{j,\alpha} W[C_{j,\alpha}] \rangle = i\,\frac{2\pi}{\kappa}\Big(\sum_{(j,\alpha)\neq(k,\beta)} q_{j,\alpha} q_{k,\beta}\,\mathrm{Lk}(C_{j,\alpha}, C_{k,\beta}) \;+\; \sum_{j,\alpha} q_{j,\alpha}^2\,\mathrm{SL}(C_{j,\alpha})\Big).$$

For **SK belts**, each physical excitation is represented by the **pair** $C_{j,+} \sqcup C_{j,-}$ with the inherited ribbon framing; absorbing charges into κ (sector-dependent normalization), the **cross-term** between S_1 and S_2 becomes

$$\exp\Big(i\,\kappa \sum_{\alpha,\beta} \mathrm{Lk}(\gamma_{1,\alpha}, \gamma_{2,\beta})\Big) \;=\; \exp\big(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)\big),$$

and the **diagonal** terms $\sum_{j,\alpha} \mathrm{SL}(\gamma_{j,\alpha})$ produce the **framing anomaly** (a pure phase fixed once the ribbon framing is chosen). This proves the formula with U_{local} absorbing contact terms and non-topological short-range pieces. ■

6.3 Non-Abelian / Berry–Chern reduction (framed Wilson algebra)

In non-Abelian Chern–Simons–type sectors (or adiabatic Berry–Chern bands), the framed Wilson loop algebra is **central–extension–like**: the product of two framed loops acquires a phase controlled by their framed linking; self-link contributes a representation-dependent shift (the well-known "framing anomaly"). Our belts implement precisely the **framed regularization** via the SK double boundary. Concretely, by non-Abelian Stokes, the belt operator

$$W[S] = \mathrm{Tr}\;\mathcal{P}\exp \iint_S \mathcal{F}$$

reduces (in these sectors) to the framed Wilson operator on ∂S. Standard loop-algebra manipulations then give

$$W[S_1]\,W[S_2] \;=\; e^{\,i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1,S_2)}\,\widetilde{W}[S_1, S_2],$$

where $\widetilde{W}$ is the ordered product with local (non-topological) renormalizations, and κ depends on the level / Berry curvature normalization and representations. The **self-link** parts appear as additive phases fixed by the framing choice, exactly as stated in the draft and its Appendix B. ■

6.4 From braiding to exchange unitary

Consider an **adiabatic exchange** of the two excitations represented by S_1, S_2. In spacetime, the exchange worldvolumes form a framed link whose topological class is measured by $\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)$. The path-ordered evolution operator restricted to the topological sector thus equals

$$U_{\mathrm{ex}} \;=\; \exp\big(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)\big)\cdot U_{\mathrm{local}},$$

matching the "What we proved" memo and the main-statement box in the draft. (Here U_{local} collects

metric-dependent contact terms and fades in the strict topological limit.)

6.5 Self-link (framing anomaly) is inevitable and physical

Because each SK belt carries two nearby boundary components, the **self-link** $\mathrm{SL}(\gamma_{j,\alpha})$ is a genuine integer determined by the chosen framed ribbon (Sec. 3). Its contribution is a constant phase per excitation, shifting under changes of framing in the **standard** way; fixing the SK ribbonization fixes that phase. This is exactly the "framing anomaly" accounted for in the framed Wilson literature and recorded in the draft's Appendix B.

6.6 Scope and objectivity

Scope. The derivation applies to **Berry–Chern / Chern–Simons–type** effective sectors; we do **not** claim a new 3+1D spin–statistics theorem. We prove a **representation-equivalence**: the exchange **phase content** equals **framed belt linking**. Non-topological scattering is absorbed into U_{local}.

Objectivity. When effects commute and records are redundantly accessible (SBS), these belt observables are **AB-fixed** and invariant under **Collapse-Lorentz** frame transforms by the SROQD theorems (imported, not re-proved). Hence the measured exchange phase is operationally well-posed.

Boxed result (Theorem C)

For framed SK belts S_1, S_2,

$$\boxed{U_{\mathrm{ex}} \;=\; \exp\left(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1,S_2)\right)\cdot U_{\mathrm{local}}\,,}$$

with κ fixed by the effective sector and **self-link** phases set by the ribbon framing. This completes the rigorous derivation of exchange/statistics from **framed belt linking**, as announced in the draft.

7. Objectivity and Frame Invariance of Belt Observables

(Imported theorems from SROQD; no new assumptions here)

We show that the belt observables introduced earlier—(i) **belt holonomy phase** $\Phi = \arg \mathrm{Tr}\, \mathcal{H}[S]$ and (ii) **framed belt linking** $\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)$—are **objective** (AB-fixed) and **frame-invariant** under the hypotheses already proven in **SROQD**: **compatible effects** (commuting or jointly measurable), and **accessible records** (e.g., SBS redundancy). We merely *apply* SROQD's theorems to these observables.

7.1 Setup: observers, effects, and records

Two adaptive observers A, B implement the same control loop γ (possibly in distinct frames) and read out:

a **phase-bin effect** E_I ("$\Phi \in I \subset \mathbb{R}/2\pi\mathbb{Z}$"), realized operationally via an interference/contrast readout recorded in the observer's trace, and

for a pair of belts S_1, S_2, an **integer-valued effect** E_k ("$\mathrm{Lk}_{\mathrm{belt}} = k$"), obtained from standard framed-Wilson interferometry (both effects are coarse-grained, hence Borel).

Assume **compatibility** of the relevant effects across observers (commuting, hence jointly measurable after Naimark dilation) and an **accessible record** (either written in the shared memory/trace or redundantly encoded in environment fragments, i.e., SBS). These are exactly the SROQD premises for cross-observer agreement.

7.2 AB-Fixedness (agreement) for belt observables

Theorem 7.1 (AB-fixedness of Φ and $\mathrm{Lk}_{\mathrm{belt}}$).

Let A, B evaluate the same belt observable (phase bin E_I or linking value E_k) under a **frame map** $\Lambda_{A\to B}$ that aligns their contexts/events and let the corresponding effects **commute**; suppose the outcome is stored in an **accessible record** (observer trace or SBS fragment). Then both observers assign **delta-certainty** to the same outcome: their post-measurement beliefs agree with probability 1.

Reason. This is a direct instance of **SROQD Theorem 4.3** ("AB-fixedness from frame map + compatibility + record"), which proves $\mathbb{P}_B(\text{mapped outcome} \mid \mathcal{F}_B) = 1$ whenever the mapped effect commutes with B's and a record is accessible. We apply the theorem to the belt-phase/bin effect E_I and the linking-value effect E_k. Hence belt holonomy readouts and framed-linking indices are **AB-fixed**. ■

Notes.
(i) For continuous Φ, we use arbitrarily fine Borel bins I; AB-fixedness holds at any fixed resolution, and in the limit of vanishing bin size whenever $W[S] \neq 0$.
(ii) SBS redundancy ensures **domain-universal objectivity**: many observers reading disjoint environment fragments converge on the same value with probability $\to 1$ as redundancy grows (SROQD Theorem 4.4).

7.3 Frame invariance (Collapse-Lorentz symmetry)

SROQD defines a **collapse interval** combining tick separation and channel-space distance and the **Collapse-Lorentz group** $\mathcal{L}$ as the set of frame maps that preserve this interval. It proves that both **AB-fixedness and incompatibility relations are invariant** under $\mathcal{L}$. Applying those results here: if A, B are related by $\Lambda \in \mathcal{L}$, the conditions for AB-fixedness of belt readouts remain satisfied in the mapped frame, hence belt holonomy and framed-linking readouts are **frame-invariant** in this operational sense.

Corollary 7.2 (Frame-robust belt readouts).
If $\Lambda \in \mathcal{L}$ maps $(t, \theta) \mapsto (t', \theta')$ while preserving collapse interval, then the event "$\Phi \in I$" (or "$\mathrm{Lk}_{\mathrm{belt}} = k$") remains AB-fixed after mapping. Thus, agreement on belt observables is **invariant under Collapse-Lorentz transforms**. ■

7.4 Necessity and failure modes (imported from SROQD)

SROQD provides counterexamples showing each assumption is **structurally essential**:

Non-commuting effects: no joint probability; AB-fixedness can fail.

Weak redundancy (no SBS): consensus across observers need not emerge.

Non-isometric frame maps (not in $\mathcal{L}$): agreement claims can be fabricated or destroyed by mapping time-like to space-like separations.
Hence our belt-level objectivity is neither assumed nor gratuitous; it is exactly the guarantee SROQD proves under its minimal hypotheses.

7.5 Consequence for this paper

Putting Sections 5–6 together with this section:

The **values** of the belt observables (variational holonomy for forces; framed linking for exchange) are fixed by the math (Thms B–C).

Their **operational objectivity** and **frame invariance** for compatible observers with accessible records are guaranteed by SROQD (AB-fixedness; Collapse-Lorentz symmetry). No new assumptions were added here.

One-line takeaway. Within the established SROQD layer, **belt holonomies and framed-linking indices are facts** (delta-certain and frame-robust) whenever effects commute and records are accessible —precisely the claim summarized in the draft and the "What we proved" memo.

8. Reductions, Cross-Checks, and Scope

8.1 Berry/Uhlmann limit (consistency check of Theorem B)

In the Abelian or pure-state regime, the surface ordering drops out and

$$\mathcal{H}[S] = \exp\Big(\iint_S \mathcal{F}\Big), \qquad \Phi = \arg \operatorname{Tr} \mathcal{H}[S] = \operatorname{Im} \iint_S \mathcal{F}.$$

The first variation reduces to the boundary contraction

$$\delta\,\Phi = \oint_{\partial\, S} \iota_{\delta x}\, \mathcal{F},$$

so the geometric (anomalous) force density along the loop is the familiar Berry/Uhlmann form,

$$F_i(\tau) = -\frac{\delta\,\Phi}{\delta x^i(\tau)} = -\,\iota_{\partial_i}\,\mathcal{F} \ \text{ on } \partial\, S,$$

exactly as recorded in the draft and its Appendix A derivation of the first-variation identity. This verifies that Theorem B collapses to textbook anomalous terms in the commuting/Abelian limit.

8.2 2+1D Chern–Simons sectors (consistency check of Theorem C)

In Abelian Chern–Simons theory, Gaussian evaluation with Poincaré-dual currents gives $\langle \prod W[C] \rangle = \exp\left(i\,\kappa \sum_{i<j} \mathrm{Lk}(C_i, C_j) + i\,\kappa \sum_i \mathrm{SL}(C_i)\right)$. Implementing SK ribbonization means each excitation is represented by the **two** framed boundary components of its belt, hence the cross-phase between two belts is

$$\exp\left(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)\right), \qquad \mathrm{Lk}_{\mathrm{belt}} := \sum_{\alpha,\beta \in \{+,-\}} \mathrm{Lk}(\gamma_{1,\alpha}, \gamma_{2,\beta}),$$

with the diagonal terms reproducing the **framing anomaly**. The draft states precisely this reduction and the ribbonized exchange unitary $U_{\mathrm{ex}} = \exp\left(i\kappa\,\mathrm{Lk}_{\mathrm{belt}}\right)\cdot U_{\mathrm{local}}$, which our detailed proof in §6 matches.

8.3 ’t Hooft double-line (ribbonization) heuristic

In large-N gauge diagrammatics, propagators are naturally drawn as **double lines**, i.e., narrow **ribbons**, so that index flow is tracked on **two** boundary strands. This standard picture is consistent with our belt formalism: SK already doubles the boundary, and the thin-belt framing (a choice of normal along the loop) coincides with the framed-Wilson regularization required to define self-link phases. Thus, “ribbonization” is not an add-on—it is the natural bookkeeping device that our **two-boundary** belts make mathematically explicit.

8.4 Observer layer (operational cross-check)

When effects commute and records are accessible (SBS redundancy), SROQD guarantees **AB-fixedness** and **frame robustness** of the belt readouts (phase bins for Φ; integer linking values). Therefore, the reductions above are not only formally consistent but also **operationally** stable across observers related by Collapse-Lorentz maps that preserve the collapse interval. We invoke these theorems without adding assumptions.

8.5 Scope (what lies outside the proven equivalence)

Our equivalence claims are **deliberately scoped** to the **geometric/adiabatic** sector (Theorem B) and to **Berry–Chern/Chern–Simons–type** exchange phases (Theorem C). Outside this domain, belts are not asserted to reproduce full dynamics:

Non-adiabatic responses / potential-gradient forces. Short-time, dissipative, or driven responses with non-geometric origins are not captured by $-\delta\,\Phi/\,\delta x$.

Inelastic channels and strong scattering. Amplitudes dominated by local interactions rather than curvature/holonomy data fall outside the holonomy–force equivalence.

Full spin–statistics in 3+1**D.** We match the **phase content** of framed Wilson/AB-type observables, but we **do not** replace the relativistic-QFT spin–statistics theorem; doing so would require additional axioms (microcausality, spectrum conditions) not introduced here. These scope boundaries are explicitly stated in the memo “What we proved” and reiterated in the draft.

Bottom line. The cross-checks (Berry/Uhlmann and 2+1D Chern–Simons) validate that our theorems reduce to known results in their natural limits, the ribbon/double-line picture aligns with the belt’s two-boundary structure, and SROQD certifies operational objectivity—while the scope clarifies what we **do not** claim beyond the geometric/topological sector.

9. Implications and “What We Proved”

9.1 Core equivalences (mathematical content)

Continuous forces ≡ belt–holonomy variation.
For any admissible control loop γ with SK belt $S(\gamma)$ and Uhlmann curvature $\mathcal{F}$,

$$F_i(\tau) \;=\; -\frac{\delta}{\delta x^i(\tau)}\,\arg\mathrm{Tr}\;\mathcal{P}\exp\iint_{S(\gamma)}\mathcal{F}.$$

Thus the geometric (Berry/Uhlmann-type) component of response is exactly the first variation of the **surface** holonomy phase. This is representation-equivalence, not a model choice.

Exchange/statistics ≡ framed belt linking.
In Berry–Chern / Chern–Simons-type sectors, for two excitations represented by framed SK belts S_1, S_2,

$$U_{\mathrm{ex}} \;=\; \exp\left(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1,S_2)\right)\cdot U_{\mathrm{local}}, \qquad \mathrm{Lk}_{\mathrm{belt}} = \sum_{\alpha,\beta\in\{+,-\}}\mathrm{Lk}(\gamma_{1,\alpha},\gamma_{2,\beta}),$$

with framing anomaly captured by the belt's self-link. Hence the exchange phase content is a **framed linking invariant** of SK ribbons.

9.2 Non-claims (explicit scope control)

No new ontology. "Belts" are not posited as new matter/fields; they are the **minimal worldsheets** the standard formalism forces upon us when SK doubling and mixed-state parallel transport are present.

No new 3+1D spin–statistics theorem. We **match** exchange phases in the usual effective/topological sectors; a full derivation of spin–statistics in 3+1D would require extra axioms (e.g., microcausality, spectrum condition) that we do not assume.

9.3 Value: inevitability and operational objectivity

Inevitability (math-first). As horizons follow from GR's equations, **belts follow** from standard ingredients: SK (two oriented boundaries), Uhlmann curvature (mixed-state geometry), and framed Wilson regularization (well-defined phases). A single unframed line cannot, in general, carry the surviving framing data in the thin-belt limit; the **two-boundary ribbon is minimal**.

Operational status. Under commuting effects and accessible records (SBS), belt readouts (holonomy phases; framed-link indices) are **AB-fixed** and **frame-invariant** under Collapse-Lorentz maps—imported theorems guaranteeing cross-observer agreement without extra postulates.

9.4 What this buys in practice

A **unified calculus**: forces from variational derivatives of surface holonomies; exchange from framed linking.

Consistency checks passed: Berry/Uhlmann limit and 2+1D Chern–Simons braiding emerge as strict reductions; 't Hooft double-line matches the ribbonization.

Clear boundary of claims: non-adiabatic, inelastic, or purely local potential-driven responses lie outside our equivalences.

One-line takeaway. Within standard quantum/open-system geometry, **belts are not an assumption—they're the mathematically inevitable carriers** of the geometric force and exchange-phase content, with observer-level objectivity secured by established results.

10. Outlook: Effective Closures and Experimental Heuristics *(no new assumptions)*

10.1 From theorems → closures (downstream, optional)

Our results already fix what is *invariantly measurable*: (i) geometric response as the first variation of the belt holonomy; (ii) exchange phase as framed belt linking. If one later wants a **computable effective model** for experiments, a light-touch *two-phase* closure (nicknamed **MBFHT**) can be adopted **downstream** (not assumed here): introduce two local phases (one per SK boundary) and write an energy functional with only terms that repackage our proven invariants,

$$\mathcal{E}[\Theta] \;=\; E_{\text{align}} + E_{\text{link}} + E_{\text{exchange}}(\,+E_{\text{drive}}),$$

where E_{link} densitizes belt holonomy, E_{exchange} encodes framed linking, and E_{align} penalizes phase misalignment between the two boundaries. This is merely a **field realization** of Theorems B–C; it adds no new physics beyond our invariants and is useful only for fitting real data.

Mapping (schematic):

Belt holonomy phase $\Phi = \arg \operatorname{Tr} \mathcal{P}\, e^{\iint_S \mathcal{F}}$ ⇝ a local "twist/curvature density" in E_{link}.

Framed belt linking Lk_{belt} ⇝ an anyonic/CS-like term in E_{exchange}.

No ontology upgrade: belts remain a *representation* of standard geometry.

10.2 Heuristic experimental signatures (illustrations, not postulates)

All signatures below are direct corollaries/diagnostics of Theorems B–C when probed adiabatically; they **do not** assume MBFHT.

Chirality-odd scattering / loop reversal test.
Drive the same closed control loop in opposite orientations (CW/CCW). By Thm. B, the geometric force density flips sign with loop orientation through $F_i(\tau) = -\delta\,\Phi/\,\delta x^i(\tau)$; any observed **odd** component isolates the holonomy contribution from potential-gradient forces.

Non-monotonic response vs. loop amplitude.
Sweep the loop size (or belt "thickness" regulator) and measure the anomalous component of the response. The phase Φ winds with enclosed curvature, producing **oscillatory / non-monotonic** trends characteristic of geometric transport, distinct from monotone dissipative backgrounds.

4π-exchange interferometry (braid-twice test).
For two excitations represented by belts S_1, S_2, implement controlled exchanges. Thm. C gives

$$U_{\text{ex}} = \exp\left(i\kappa\, \text{Lk}_{\text{belt}}(S_1, S_2)\right) \cdot U_{\text{local}},$$

so interferometric contrast is periodic in the framed-linking integer. Setups where a **double exchange** is needed to return the phase (effective 4π periodicity) provide a crisp topological check of framed-linking control.

Framing toggle / self-link calibration.
Change the belt's normal-field twist by one unit (thin-belt framing). A constant phase offset—**the**

framing anomaly—must shift accordingly while cross-link phases between distinct belts remain fixed. This separates self-link (calibration) from mutual linking (signal).

10.3 Protocol hygiene (operational objectivity)

To make belt readouts **observer-robust** in practice, implement the SROQD conditions: use **compatible effects** (commuting readouts) and ensure **accessible records** (shared memory or SBS redundancy). Under these hypotheses, phase bins for Φ and integer $\mathrm{Lk}_{\mathrm{belt}}$ outcomes are **AB-fixed** and remain invariant under **Collapse-Lorentz** frame maps that preserve the collapse interval—so different labs/frames agree on the same belt facts.

10.4 Scope reminder

The above heuristics probe exactly what we proved; they **do not** claim universality across non-adiabatic, strongly inelastic, or purely potential-gradient regimes (outside the geometric/topological sector we scoped).

Takeaway. Our theorems already pin down the *invariants*. If one wants predictions "with knobs," an optional two-phase closure can parametrize those invariants for data fitting. Either way, **belts are inevitable** as the minimal worldsheets carrying geometric-force and exchange-phase content; SROQD guarantees their operational objectivity when measured properly.

Appendix A — First-Variation Formula for $\arg \mathrm{Tr}\ \mathcal{H}[S]$

We derive the first-variation of the **belt holonomy phase**

$$\Phi := \arg \mathrm{Tr}\ \mathcal{H}[S], \qquad \mathcal{H}[S] := \mathcal{P}\exp \iint_S \mathcal{F},$$

for a smooth family of belts $S = S_\sigma$ with moving boundary $\partial S_\sigma = \gamma_{+,\sigma} \sqcup \gamma_{-,\sigma}$. The result used informally in the draft (Appendix A) is justified here by an ordered–exponential (Duhamel/Volterra) calculus plus Cartan's identity; we also state sufficient analytic hypotheses and gauge-covariance.

A.0 Standing hypotheses and notation

Geometry. Θ is a smooth manifold; the SK base is the disjoint union $\mathrm{SK} = \{+, -\}$. Belts $S_\sigma \subset \Theta \times \mathrm{SK}$ are smooth immersed compact oriented surfaces with boundary $\partial S_\sigma = \gamma_{+,\sigma} \sqcup \gamma_{-,\sigma}$. We fix a smooth family of embeddings $\iota_\sigma : A \to \Theta \times \mathrm{SK}$ with $A := S^1 \times I$ ($I = [-1/2, 1/2]$) so that $S_\sigma = \iota_\sigma(A)$.

Connection/curvature. $\mathcal{A}$ is the (possibly non-Abelian) Uhlmann/Bures connection on the pulled-back purification bundle; $\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}$ its curvature 2-form. Non-Abelian Stokes is used in surface form.

Regularity (sufficient).
(R1) ι_σ is C^1 in σ and C^∞ on A; $\mathcal{A}$ is C^1 and $\mathcal{F}$ is continuous on a neighborhood of $\bigcup_\sigma S_\sigma$.
(R2) We evaluate in a finite-dimensional representation so $\mathcal{H}[S_\sigma]$ is a trace-class matrix and $W(\sigma) := \mathrm{Tr}\,\mathcal{H}[S_\sigma]$ is C^1.
(R3) $W(\sigma) \neq 0$ on the variation interval, so $\Phi(\sigma) = \arg W(\sigma)$ is well-defined and $d\Phi = \mathrm{Im}(dW / W)$.

These mirror the draft's Appendix-A assumptions.

We write $V := \partial_\sigma \iota_\sigma$ for the deformation vector field along S_σ, and $\delta x := V|_{\partial S_\sigma}$ the induced **boundary displacement**. The interior product (contraction) with a vector is denoted by $\iota_{(\cdot)}$.

A.1 Duhamel/Volterra formula for surface-ordered exponentials

Let $\mathcal{K}_\sigma := \iota_\sigma^* \mathcal{F}$ be the pulled-back curvature on the fixed parameter domain A. The surface-ordered exponential

$$\mathcal{U}_\sigma := \mathcal{P}\exp \iint_A \mathcal{K}_\sigma \quad (\text{so that } \mathcal{H}[S_\sigma] = \mathcal{U}_\sigma)$$

has first variation

$$\delta \mathcal{U}_\sigma = \iint_A \mathcal{P}\Big[\mathcal{U}_\sigma^{>(\xi)}\, \delta \mathcal{K}_\sigma(\xi)\, \mathcal{U}_\sigma^{<(\xi)}\Big] d\xi + (\text{higher nested commutators}), \qquad (A.1)$$

where $\xi \in A$ is a surface-order parameter and $\mathcal{U}_\sigma^{>(\xi)}$ (resp. $^{<(\xi)}$) is the ordered factor "after" (resp. "before") ξ. Equation (A.1) is the surface analogue of Duhamel's formula and is the precise backbone behind the draft's "Duhamel/Volterra series" wording.

The variation of the pulled-back curvature is given by **Cartan's identity** for the Lie derivative of a form under the domain deformation:

$$\delta \mathcal{K}_\sigma = \frac{d}{d\sigma}\, \iota_\sigma^* \mathcal{F} = \iota_\sigma^* \big(\mathcal{L}_V \mathcal{F}\big) = \iota_\sigma^* \big(d\, \iota_V \mathcal{F} + \iota_V\, d\mathcal{F}\big). \qquad (A.2)$$

Using the **Bianchi identity** $D\mathcal{F} := d\mathcal{F} + [\mathcal{A}, \mathcal{F}] = 0$, the interior term $\iota_V d\mathcal{F} = -\iota_V [\mathcal{A}, \mathcal{F}]$ can be absorbed by the nested-commutator corrections that appear automatically in (A.1). This is the non-Abelian content of the variation and is exactly the "commutator control" announced in the draft.

A.2 Boundary reduction: the $\iota_{\delta x}\, \mathcal{F}$ term

Insert (A.2) into (A.1). The piece with $d(\iota_V \mathcal{F})$ is reduced by Stokes on the **fixed parameter domain** A (with moving image S_σ):

$$\iint_A \mathcal{P}\big[\cdots d(\iota_V \mathcal{F}) \cdots\big] = \oint_{\partial A} \mathcal{P}\big[\cdots \iota_V \mathcal{F} \cdots\big] \mapsto \oint_{\partial S_\sigma} \mathcal{P}_{\partial S}\big[\iota_{\delta x} \mathcal{F}\big], \qquad (A.3)$$

where $\mathcal{P}_{\partial S}[\,\cdot\,]$ denotes the boundary-ordered insertion obtained after transporting the Lie-algebra element from its interior evaluation point to the boundary segment according to the surface ordering. Concretely, for a point $p \in \partial S_\sigma$, the insertion reads

$$\mathcal{P}_{\partial S}\big[\iota_{\delta x} \mathcal{F}\big](p) := \mathrm{Ad}\left(\mathcal{U}_\sigma^{>(p)}\right)\big(\iota_{\delta x} \mathcal{F}\big),$$

with $\mathrm{Ad}(U)(X) = U^{-1} X U$. (Any equivalent "parallel-transport-to-boundary" convention yields the same conjugacy class and hence the same trace/phase.) This makes precise the draft's boundary contraction term $\iota_{\delta x}\, \mathcal{F}$.

The remaining contribution from $\iota_V d\mathcal{F}$ organizes into a boundary-ordered **commutator series** (the "Volterra tails"). Their covariant sum is a class function under gauge transformations and vanishes identically in the Abelian/pure-state limit (Section A.5).

A.3 First-variation of $W(\sigma) = \operatorname{Tr} \mathcal{H}[S_\sigma]$ and of its phase

Taking traces in (A.1)–(A.3) gives

$$\delta W = \oint_{\partial S_\sigma} \left\langle \mathcal{P}_{\partial S}\left[\iota_{\delta x} \mathcal{F}\right] \right\rangle + \text{(explicit nested commutators with total trace)}, \qquad \text{(A.4)}$$

where $\langle \cdot \rangle = \operatorname{Tr}(\cdot)$ in the fixed representation. Whenever $W \neq 0$, the phase $\Phi = \arg W$ varies as

$$\delta \Phi = \operatorname{Im} \frac{\delta W}{W} = \operatorname{Im} \left(\frac{\oint_{\partial S_\sigma} \left\langle \mathcal{P}_{\partial S}\left[\iota_{\delta x} \mathcal{F}\right] \right\rangle}{\langle \mathcal{U}_\sigma \rangle} \right) + \operatorname{Im} \left(\frac{\text{commutator tails}}{\langle \mathcal{U}_\sigma \rangle} \right). \qquad \text{(A.5)}$$

It is often convenient—and equivalent for our purposes—to package the transport and the commutator series into a single **gauge-covariant boundary functional** $\mathfrak{B}_{S_\sigma}[\delta x; \mathcal{F}]$ so that

$$\boxed{\delta \Phi = \oint_{\partial S_\sigma} \left\langle \mathfrak{B}_{S_\sigma}[\delta x; \mathcal{F}] \right\rangle, \quad \mathfrak{B}_{S_\sigma}[\delta x; \mathcal{F}] = \mathcal{P}_{\partial S}\left[\iota_{\delta x} \mathcal{F}\right] + \text{(ordered commutator series)}.} \qquad \text{(A.6)}$$

Equation (A.6) is the rigorous version of the draft's succinct formula "
$\delta \arg \operatorname{Tr} \mathcal{H}[S] = \oint \langle \iota_{\delta x} \mathcal{F} \rangle + \text{(commutators)}$" and is the only input needed for Theorem B (force = variational derivative).

A.4 Gauge covariance and reparametrization invariance

Gauge covariance. Under a smooth gauge transform g: $\mathcal{F} \mapsto g^{-1} \mathcal{F} g$ and $\mathcal{U}_\sigma \mapsto G_+^{-1} \mathcal{U}_\sigma G_-$. Each boundary insertion in (A.6) is conjugated accordingly, so $\langle \cdot \rangle$ and Φ are invariant (class functions). This matches the draft's gauge-covariance statement.

Reparametrizations. Changing the parameterization on A or ∂S_σ leaves (A.6) invariant because $\mathcal{F}$ is a 2-form and $\iota_{\delta x} \mathcal{F}$ is a contraction with a geometric boundary vector field, not with a coordinate artifact.

A.5 Abelian/pure-state reduction

If the structure group effectively commutes along S_σ (Abelian or pure-state reduction), ordering is immaterial: $\mathcal{U}_\sigma = \exp\left(\iint_{S_\sigma} \mathcal{F} \right)$. Then (A.4) collapses to

$$\delta W = W \cdot \iint_{S_\sigma} d(\iota_V \mathcal{F}) = W \cdot \oint_{\partial S_\sigma} \iota_{\delta x} \mathcal{F},$$

so the commutator series vanishes and

$$\boxed{\delta \Phi = \oint_{\partial S_\sigma} \iota_{\delta x} \mathcal{F} \qquad \text{(Abelian / pure-state)}.} \qquad \text{(A.7)}$$

This is precisely the "textbook" Berry/Uhlmann first-variation identity recovered in the draft.

A.6 Functional derivative and Theorem B (for completeness)

Write the loop boundary in local coordinates $x^i(\tau)$ and consider a Gâteaux boundary deformation $\delta x^i(\tau)$. From (A.6),

$$\delta\,\Phi = \int_{S^1} \left\langle \mathfrak{B}_{S_\sigma}[\partial_i; \mathcal{F}] \right\rangle \delta x^i(\tau)\, d\tau \quad \Longrightarrow \quad \frac{\delta\,\Phi}{\delta x^i(\tau)} = \left\langle \mathfrak{B}_{S_\sigma}[\partial_i; \mathcal{F}] \right\rangle,$$

so the **geometric force density** is

$$\boxed{F_i(\tau) \;=\; -\,\frac{\delta\,\Phi}{\delta x^i(\tau)}\,,}$$

exactly as stated and proved in Section 5, with (A.7) giving the Berry/Uhlmann form in the Abelian limit.

A.7 Analytic justifications (sufficiency)

(J1) **Differentiation under trace and integral.** Continuity of $\mathcal{F}$ on $\bigcup_\sigma S_\sigma$ and C^1 dependence on σ ensure dominated convergence for the Volterra series on compact A; the ordered products converge uniformly on compacta in a finite-dimensional representation.

(J2) **Non-vanishing of W.** If $W(\sigma_0) = 0$ at isolated points, one can work with a branch cut; the variation formula applies on each connected component where $W \neq 0$.

(J3) **Shape derivative framework.** Using a fixed parameter domain A and moving embeddings ι_σ converts domain variations to **pullback** variations, so Cartan's identity applies directly (no extra boundary terms beyond those produced by $d(\iota_V\,\mathcal{F})$).

(J4) **Gauge covariance.** Because (A.6) is built from conjugacy-invariant data (trace of conjugates), the phase is gauge-invariant; this matches the draft's gauge-covariance remarks around the non-Abelian Stokes formulation.

A.8 Summary (connection to the draft)

Equations (A.6)–(A.7) are the rigorous versions of the draft's Appendix-A line:

$$\delta \arg \mathrm{Tr}\, \mathcal{H}[S] \;=\; \oint_{\partial S} \langle\, \iota_{\delta x}\, \mathcal{F}\, \rangle \;+\; (\text{commutator terms}),$$

with explicit ordered-transport bookkeeping and hypotheses spelled out; they are exactly what powers **Theorem B (forces = variational holonomy)** in the main text and the Berry/Uhlmann reduction in §8.1.

Cross-reference. For reader orientation, the informal statement and its use appear in the draft's Appendix A and Theorem B sections.

Appendix B — Framed Belt Linking and Exchange

We derive the exchange unitary from **framed Wilson regularization** applied to **SK belts** and make the **level/normalization** explicit. The result exactly matches the statement used in the main text and the

draft's Appendix B: in Berry–Chern / Chern–Simons–type sectors the exchange phase is a **framed linking invariant** of the two belts, with additive **self-link (framing) phases** and a sector-dependent **level** κ.

B.0 Standing setup and notation

A **belt** $S \subset \Theta \times \mathrm{SK}$ is a smooth immersed worldsheet with two oriented boundary curves $\partial S = \gamma_+ \sqcup \gamma_-$ (Sec. 3). Its holonomy is

$$\mathcal{H}[S] = \mathcal{P}\exp \iint_S \mathcal{F}, \qquad W[S] = \mathrm{Tr}\ \mathcal{H}[S].$$

In topological/adiabatic sectors, $W[S]$ is represented by a **framed Wilson operator** living on ∂S. The **framing** comes from the thin-belt limit; it is a nowhere-vanishing normal along each boundary.

For two belts S_1, S_2 with boundaries $\gamma_{j,\pm}$, define the **framed belt linking**

$$\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) \ = \sum_{\alpha,\beta \in \{+,-\}} \mathrm{Lk}(\gamma_{1,\alpha}, \gamma_{2,\beta}).$$

Self-link of a *framed* loop γ is denoted $\mathrm{SL}(\gamma) \in \mathbb{Z}$.

B.1 From belt holonomies to framed Wilsons (SK ribbonization)

By the non-Abelian Stokes map used in the draft, the surface operator $W[S]$ reduces in the topological/adiabatic regime to a framed Wilson object on ∂S. The SK belt already provides the **two nearby boundary components** (the "ribbon"), which is the standard regularization required for well-defined phases in Chern–Simons-type theories. Thus the belt itself furnishes the **framing regulator**; the self-link phase that survives the thin-belt limit is precisely the **framing anomaly** term.

B.2 Abelian Chern–Simons derivation (complete, with level normalization)

Work in 2+1D with Abelian Chern–Simons action at **level** $k \neq 0$ (we keep the normalizations explicit):

$$S_{\mathrm{CS}}[A] \ = \ \frac{k}{4\pi} \int_{\mathbb{R}^{1,2}} A \wedge dA, \qquad A \in \Omega^1.$$

A framed Wilson operator of **charge** $q \in \mathbb{R}$ along an oriented loop C is $W_q[C] = \exp\left(iq \oint_C A\right)$.

Let J_C be the Poincaré-dual 2-form current of C so that $\int \eta \wedge J_C = \oint_C \eta$. For a set of (pairwise disjoint) framed loops $\{C_a\}$ with charges $\{q_a\}$, define $J = \sum_a q_a J_{C_a}$. The generating functional is Gaussian:

$$\mathcal{Z}[J] = \int \mathcal{D}A \, \exp\left\{ i \frac{k}{4\pi} \int A \wedge dA + i \int A \wedge J \right\}.$$

Completing the square with the Green operator d^{-1} on coexact 2-forms gives the exact value

$$\log \frac{\mathcal{Z}[J]}{\mathcal{Z}[0]} \ = \ \frac{i\,2\pi}{k} \int J \wedge d^{-1} J \,.$$

Using the standard distributional identity (Hopf pairing),

$$\int J_C \wedge d^{-1} J_{C'} \;=\; \mathrm{Lk}(C, C'), \qquad \int J_C \wedge d^{-1} J_C \;=\; \mathrm{SL}(C) \quad \text{(framed)},$$

we obtain

$$\left\langle \prod_a W_{q_a}[C_a] \right\rangle = \exp\left\{ i \frac{2\pi}{k} \Big(\sum_{a<b} q_a q_b \,\mathrm{Lk}(C_a, C_b) \;+\; \frac{1}{2} \sum_a q_a^2 \,\mathrm{SL}(C_a) \Big) \right\}.$$

Level normalization. Comparing with the main-text form $\exp\big(i\,\kappa\,\mathrm{Lk}\big)$, the **effective** κ between two charged loops is

$$\boxed{\kappa_{\mathrm{eff}} = \frac{2\pi}{k}\, q_a q_b}.$$

For **SK belts**, each excitation contributes two boundary loops $C_{j,\pm} = \gamma_{j,\pm}$. Summing over the four mutual pairs yields exactly

$$\exp\Big(i\,\kappa_{\mathrm{eff}}\,\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) \Big),$$

while the diagonal terms $\propto \sum_{j,\alpha} q_{j,\alpha}^2\,\mathrm{SL}(\gamma_{j,\alpha})$ contribute the **self-link (framing) anomaly**—a belt-intrinsic phase fixed once the SK framing is fixed. This is the draft's Appendix-B statement, now with the level normalization explicit. ■

B.3 Non-Abelian / Berry–Chern reduction (representation data and framing)

In non-Abelian Chern–Simons–type sectors or adiabatic Berry–Chern bands, replace q by the representation data (e.g., a quadratic Casimir factor in the appropriate limit), and interpret $W[S] = \mathrm{Tr}_R\, \mathcal{P}\exp \iint_S \mathcal{F}$ as the framed Wilson operator in representation R on ∂S. The **product** of two framed Wilsons picks up a central phase governed by **framed linking**, while **self-link** induces a representation-dependent framing phase; all **local** (metric-dependent) contributions are absorbed into U_{local}. In this reduction, the exchange unitary has the same functional form:

$$U_{\mathrm{ex}} \;=\; \exp\big(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) \big) \cdot U_{\mathrm{local}},$$

with κ set by the sector's level/representation normalization. This is the non-Abelian counterpart of the Abelian calculation above and is the **ribbon** version of the framed Wilson algebra quoted in the draft.

> **Remark on level conventions.** Different conventions (e.g., k vs k + shift) amount to a redefinition of κ and the self-link calibration; our main-text statements purposely package all such choices into the single real constant κ and the framing-fixed self-link phases, exactly as framed in the draft.

B.4 SK specifics: why "belt linking" (not just curve linking)

Because each excitation is represented by a **belt** (two nearby boundary components with inherited framing), the mutual topological content is

$$\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) = \sum_{\alpha,\beta} \mathrm{Lk}(\gamma_{1,\alpha}, \gamma_{2,\beta}),$$

which is precisely what the Gaussian evaluation (Abelian case) or framed Wilson algebra (non-Abelian

case) sums over. A single unframed curve would **miss** the finite self-link term and, in general, cannot reproduce the correct exchange phase—hence the necessity of the ribbon regularization supplied by SK belts.

B.5 Framing anomaly: calibration and invariance

Changing the belt's framing by one unit shifts $\mathrm{SL}(\gamma_{j,\alpha}) \mapsto \mathrm{SL}(\gamma_{j,\alpha}) \pm 1$, producing a **constant phase** that depends only on the chosen framing convention; **mutual** linking $\mathrm{Lk}(\gamma_{1,\alpha}, \gamma_{2,\beta})$ is unaffected. Thus:

Calibrate once (pick the SK thin-belt framing) → self-link phases fixed.

Physics of **exchange** resides in the **mutual** framed linking captured by $\mathrm{Lk}_{\mathrm{belt}}$.
This is exactly how the draft encodes "self-link (framing) contributes additive phases."

B.6 Boxed result (exchange from framed belt linking)

Putting B.2–B.5 together:

$$U_{\mathrm{ex}} \;=\; \exp\left(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)\right)\cdot U_{\mathrm{local}}, \qquad \kappa \text{ fixed by level/representation (e.g., } \kappa = \tfrac{2\pi}{k}\,q_1 q_2 \text{ in Abelian CS).}$$

Self-link (framing) produces additive constant phases fixed by the SK ribbon framing; U_{local} absorbs non-topological contact terms. This is precisely the statement used in the main text and the draft's Appendix B.

B.7 Relation to the paper's claims

What we proved. Exchange/statistics **phase content** equals **framed belt linking** (no new ontology; level is a sector constant), in line with the memo.

Operational objectivity (pointer). When readouts are compatible and records accessible, these belt-level phases are AB-fixed and frame-robust by SROQD (cited earlier), but no such assumptions were needed for the *derivation* of the linking formula itself.

This completes Appendix B's proof pack for framed linking and exchange, including level normalization and framing-anomaly handling.

Appendix C — Operator-Algebraic and Process-Tensor Recast

We recast belts and their holonomies in (i) **von Neumann algebraic** language and (ii) the **process-tensor/quantum-comb** formalism. We then **map them to SROQD's filtrations** to justify operational measurability (AB-fixedness; Collapse-Lorentz invariance) with *no new assumptions*.

C.0 Standing setup

World and memory Hilbert spaces $\mathcal{H}_W, \mathcal{H}_M$ (separable), total $\mathcal{H} = \mathcal{H}_W \otimes \mathcal{H}_M$. Instruments $\{\mathcal{M}_{\theta,a}\}_{a\in\Phi}$ (CP, $\sum_a$ TP), adaptive policy $\theta_t = \pi_t(A_1, \ldots, A_{t-1})$. Observer trace produces an increasing filtration. (Same layer as SROQD.)

SK contour provides two oriented branches; a loop γ lifts to $\gamma_{\pm}$ and spans a **belt** $S(\gamma) \subset \Theta \times \mathrm{SK}$. Holonomy:

$$\mathcal{H}[S] = \mathcal{P}\exp \iint_S \mathcal{F}, \qquad W[S] = \mathrm{Tr}\ \mathcal{H}[S], \qquad \Phi = \arg W[S].$$

(Uhlmann curvature $\mathcal{F}$; non-Abelian Stokes.)

C.1 Operator-algebraic recast (von Neumann expectations)

C.1.1 Algebras, states, instruments, filtration

Let $\mathcal{N}_W \subset \mathcal{B}(\mathcal{H}_W)$, $\mathcal{N}_M \subset \mathcal{B}(\mathcal{H}_M)$ be von Neumann algebras; $\mathcal{N} := \mathcal{N}_W \bar{\otimes} \mathcal{N}_M$. Fix a **normal state** ω on $\mathcal{N}$. An instrument at context θ is the normal CP map

$$\mathfrak{I}_\theta(X) = \sum_{a \in \Phi} \mathcal{M}_{\theta,a}(X) \otimes |a\rangle\langle a|_M.$$

The observer's **filtration** is the increasing tower of von Neumann subalgebras

$$\mathcal{F}_t := \mathrm{vN}\big(\text{pointer projectors for } A_1, \ldots, A_t\big) \subset \mathcal{N}_M,$$

with **conditional expectations** $\mathbb{E}_t : \mathcal{N} \to \mathcal{F}_t$. (SROQD §6.1.)

C.1.2 Belt holonomy as a normal functional of a Wilson–belt operator

Let $\mathcal{U}_S$ denote a (surface-ordered) parallel-transport **unitary** on a purification fiber implementing $\mathcal{H}[S]$ (non-Abelian Stokes). Embed $\mathcal{U}_S$ into $\mathcal{N}$ by a Stinespring isometry V for the Uhlmann bundle pullback; define the **Wilson–belt operator**

$$\mathbf{W}[S] := V^\dagger(\mathcal{U}_S \otimes \mathbf{1})V \in \mathcal{N}.$$

Then the belt functional used in the main text is the **normal expectation**

$$W[S] = \omega(\mathbf{W}[S]), \qquad \Phi = \arg \omega(\mathbf{W}[S]).$$

Gauge covariance. A gauge transform acts as an **inner automorphism** $\alpha_g(X) = U_g^\dagger X U_g$ on the pulled-back bundle. Since ω is a normal functional and we take a **class function** (phase of a trace/expectation), Φ is invariant under α_g.

Framing and SK. The two SK boundaries make $\mathbf{W}[S]$ a **framed** object; the thin-belt limit keeps a self-link (framing) contribution, not representable by a single unframed line operator. (Matches Theorem A.)

C.1.3 Measurable belt effects and AB-fixedness

Define Borel **phase-bin effects** on $\mathbb{S}^1$:

$$E_I := \mathbf{1}_{\{\arg \omega(\mathbf{W}[S]) \in I\}} \in \mathcal{F}_{t+1},$$

operationally implemented by an interferometric readout whose pointer is written at tick $t{+}1$. Similarly, for two belts S_1, S_2, define an **integer-valued effect** E_k registering $\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) = k$ via a framed-Wilson interferometer (pointer in $\mathcal{F}_{t+1}$). By SROQD:

(**Delta-certainty / latching**): For any $F \in \mathcal{F}_t$, $\mathbb{E}_t(F) = F$, hence once recorded, the event is **fixed** for the observer. (Conditional expectation fixed-point.)

(**AB-fixedness**): If two observers use **commuting effects** and share an **accessible record** (SBS), the mapped effects yield agreement with probability 1. (Apply SROQD Theorem on AB-fixedness.)

(**Collapse-Lorentz invariance**): Agreement/incompatibility are preserved under frame maps that preserve the collapse interval, so belt readouts are **frame-robust**.

Thus Φ and $\mathrm{Lk}_{\mathrm{belt}}$ correspond to **effects in the filtration** and are operationally measurable without new axioms.

C.2 Process-tensor / quantum-comb recast (CTP/SK form)

C.2.1 Process tensor with SK legs

Let $\mathcal{T}_{0:T}$ be the **process tensor** (Choi operator) that, for a sequence of instruments, returns outcome probabilities. On the SK contour, each tick contributes a **forward** and **backward** leg; the global object is a **CTP comb T** on doubled spaces. (SROQD §6.2 sketch).

C.2.2 Belts as CTP surface contractions

A control loop γ picks instruments with $\theta_t = \pi_t(\cdot)$. The SK belt $S(\gamma)$ is represented at the comb level by inserting a **surface-ordered curvature functional** $\mathfrak{S}[\mathcal{F}; S]$ that couples the forward/back legs along the annulus spanned by $\gamma_{\pm}$:

$$\mathfrak{S}[\mathcal{F}; S] \;\rightsquigarrow\; \text{CTP influence functional } \mathcal{I}_S \;=\; \mathcal{P}\exp \iint_S \mathcal{F},$$

so that the **Wilson–belt amplitude** is the **link product** (contraction)

$$W[S] \;=\; \mathrm{Tr}\,[\mathbf{T} \star \mathcal{I}_S], \qquad \Phi = \arg W[S],$$

with $\star$ the comb link (Choi) composition. In Abelian/pure-state reduction this equals the familiar CTP influence phase; in general it is the SK **surface** holonomy.

C.2.3 Variations (Theorem B) and framed linking (Theorem C) in the comb

First variation. A boundary deformation δx of S induces

$$\delta\,\mathcal{I}_S \;=\; \oint_{\partial S} \iota_{\delta x}\,\mathcal{F} \;+\; (\text{commutator tails}),$$

hence

$$\delta\,\Phi \;=\; \mathrm{Im}\,\frac{\mathrm{Tr}\big[\mathbf{T}\star\delta\,\mathcal{I}_S\big]}{\mathrm{Tr}\big[\mathbf{T}\star\mathcal{I}_S\big]} \quad\Rightarrow\quad F_i(\tau) \;=\; -\,\frac{\delta\,\Phi}{\delta x^i(\tau)}.$$

This is the **comb** version of Appendix-A's formula.

Exchange as framed linking. Insert two belt functionals $\mathcal{I}_{S_1}, \mathcal{I}_{S_2}$. In **Berry–Chern / Chern–Simons-type** sectors, Gaussian/CS reduction of the CTP action yields a central phase

$$\mathrm{Tr}\left[\mathbf{T}\star\mathcal{I}_{S_1}\star\mathcal{I}_{S_2}\right] \;=\; e^{\,i\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1,S_2)}\,\mathrm{Tr}\left[\mathbf{T}\star\widetilde{\mathcal{I}}\right],$$

with **self-link** providing the framing anomaly. This reproduces the framed-Wilson algebra in comb language (Theorem C).

C.3 Operational measurability via SROQD filtrations

Effects live in the filtration. Phase-bin and linking-value readouts are implemented as POVMs whose **pointer projectors** lie in $\mathcal{F}_{t+1}$; hence they are **measurable** with respect to the observer's past. (SROQD operator-algebraic §6.1.)

AB-fixedness and invariance. With commuting effects and accessible records (SBS), SROQD proves **delta-certainty** and **frame invariance** (Collapse-Lorentz) for mapped outcomes—so **belt holonomy phases** and **framed linking indices** are **objective** observables in this operational sense.

Scope control. As summarized in the "What we proved" memo, this guarantees objectivity **only** for the geometric/topological content (holonomy variation; framed linking), not for non-adiabatic or inelastic sectors.

C.4 Summary

A belt holonomy is the **normal expectation** $W[S] = \omega(\mathbf{W}[S])$ of a **Wilson–belt operator** in a von Neumann algebra; its phase Φ is gauge-invariant (class function).

In the **process-tensor** picture, $W[S]$ is the **CTP influence phase** obtained by contracting the process comb with a **surface-ordered** curvature insertion; its variation yields Theorem B, and pairs of insertions yield Theorem C with **framed belt linking** and **self-link** anomaly.

Via SROQD's filtration/conditional-expectation machinery, the corresponding **effects** are measurable and **AB-fixed**; their agreement is **Collapse-Lorentz invariant**—no new assumptions added.

This completes the algebraic/comb recast and its operational justification.

Appendix D — Rectifiable Two-Chains and Regularity

(existence of spanning belts; measurability and convergence of $\mathcal{H}[S]$)

We supply the geometric-measure-theoretic underpinnings used implicitly in Theorem A and in the variation calculus of Appendix A: (i) **existence** of smooth/rectifiable **belts** that span the two SK boundaries for any admissible loop; (ii) **well-posedness** and **stability** (measurability & convergence) of the belt holonomy

$$\mathcal{H}[S] \;=\; \mathcal{P}\exp \iint_S \mathcal{F}, \qquad W[S] = \operatorname{Tr} \mathcal{H}[S], \qquad \Phi = \arg W[S].$$

The only inputs are the SK doubling, the Uhlmann bundle/curvature, and non-Abelian Stokes as formulated in the draft.

D.0 Standing assumptions

Θ a smooth, second-countable manifold endowed when needed with a smooth Riemannian metric g.

A closed control loop $\gamma \in C^1(S^1, \Theta)$. SK doubling yields two oriented boundary curves $\gamma_{\pm} \subset \Theta \times \mathrm{SK}$.

Uhlmann connection $\mathcal{A}$ with curvature $\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}$, pulled back to $\Theta \times \mathrm{SK}$. The belt holonomy is defined by the surface-ordered exponential; non-Abelian Stokes is used exactly as in the draft.

D.1 Smooth belts always exist (geodesic normal ribbon)

Theorem D.1 (smooth immersed belt).

Let $\gamma \in C^1(S^1, \Theta)$. Then there exists $\varepsilon_0 > 0$ and, for each $\varepsilon \in (0, \varepsilon_0)$, a smooth immersed compact oriented surface $S_\varepsilon(\gamma, \nu) \subset \Theta \times \mathrm{SK}$ with boundary $\partial S_\varepsilon = \gamma_+ \sqcup \gamma_-$.

Construction. Equip Θ with g. By the tubular-neighborhood theorem, there exists a nowhere-vanishing smooth normal field $\nu(\sigma) \perp \dot{\gamma}(\sigma)$. For small $\varepsilon > 0$ define offset loops

$$\gamma_\pm^\varepsilon(\sigma) := \exp_{\gamma(\sigma)}\left(\pm \frac{\varepsilon}{2}\, \nu(\sigma)\right).$$

The **geodesic normal ribbon** in Θ is the image of $\iota_\Theta(\sigma, t) = \exp_{\gamma(\sigma)}(t\,\nu(\sigma))$, $t \in [-\varepsilon/2, \varepsilon/2]$. Lift to $\Theta \times \mathrm{SK}$ by $\iota(\sigma, t) = (\iota_\Theta(\sigma, t), \mathrm{br}(t))$ with $\mathrm{br}(t) \in \{+, -\}$ equal to $\mathrm{sign}(t)$ and $\mathrm{br}(\pm\varepsilon/2) = \pm$. The image $S_\varepsilon(\gamma, \nu) = \iota(S^1 \times [-\varepsilon/2, \varepsilon/2])$ is a smooth immersed belt and $\partial S_\varepsilon = \gamma_+ \sqcup \gamma_-$. ■

Remarks.

Connectivity is **not** required: S_ε is an immersed ribbon in $\Theta \times \mathrm{SK}$ with exactly the two SK boundaries.

This is the explicit construction used in Theorem A of the draft (there as a proof sketch; here as a full statement).

D.2 Rectifiable belts for low-regularity boundaries

The previous theorem assumes $\gamma \in C^1$. We extend to loops of **finite length** (Lipschitz).

Theorem D.2 (rectifiable belt).

If γ is Lipschitz, then for every sufficiently small $\varepsilon > 0$ there exists a **rectifiable** 2-current T_ε in $\Theta \times \mathrm{SK}$ with finite mass such that $\partial T_\varepsilon = \gamma_+ \sqcup \gamma_-$. Moreover, T_ε can be approximated in mass by smooth immersed belts $S_\varepsilon^{(n)}$ with the same boundary.

Proof (outline). A Lipschitz γ admits an a.e. unit normal ν along almost every point via metric projections in a tubular neighborhood. The offset curves $\gamma_\pm^\varepsilon$ are Lipschitz and disjoint for small ε. The **annulus** between γ_-^ε and γ_+^ε inside the tubular neighborhood defines a rectifiable 2-current T_ε^Θ in Θ with $\partial T_\varepsilon^\Theta = \gamma_+^\varepsilon - \gamma_-^\varepsilon$. Lifting to $\Theta \times \mathrm{SK}$ and letting the thickness direction play the SK label yields a rectifiable belt T_ε with $\partial T_\varepsilon = \gamma_+ \sqcup \gamma_-$. Standard smoothing of Lipschitz maps in charts gives smooth immersed belts $S_\varepsilon^{(n)} \to T_\varepsilon$ in mass. ■

Use. This rectifiable setting is the one implicitly invoked when we speak of "rectifiable 2-chains" in the main text.

D.3 Existence and measurability of $\mathcal{H}[S]$

Lemma D.3 (existence of $\mathcal{H}[S]$ on rectifiable belts).

Let S be a smooth belt or a rectifiable belt approximated in mass by smooth immersed belts $S^{(n)} \to S$. Assume $\mathcal{F}$ is continuous on a neighborhood of $\mathrm{supp}\, S$ and bounded in operator norm in a fixed finite-

dimensional representation. Then the sequence of surface-ordered exponentials $\mathcal{H}[S^{(n)}] = \mathcal{P}\exp\iint_{S^{(n)}} \mathcal{F}$ **converges** (in operator norm) to a limit that depends only on S; define this as $\mathcal{H}[S]$.

Sketch. Use the Duhamel/Volterra series on each $S^{(n)}$ (Appendix A) and dominated convergence: bounded $\|\mathcal{F}\|$ and bounded area give uniform control on all iterated integral terms. The Stokes-covariant commutator bookkeeping from Appendix A ensures a common bound independent of n. ■

Corollary D.4 (measurability).

If $\gamma \mapsto S(\gamma)$ is Borel (e.g., geodesic-normal ribbon with a Borel choice of ν), then $\gamma \mapsto W[S(\gamma)] = \mathrm{Tr}\,\mathcal{H}[S(\gamma)]$ and $\gamma \mapsto \Phi(\gamma)$ are Borel. Consequently, **phase-bin** effects $E_I = \mathbf{1}_{\{\Phi \in I\}}$ live in the observer's filtration (pointer algebra), as used in §7 and Appendix C.

D.4 Stability: continuity under surface and curvature limits

Let $S_n \to S$ in the sense of **mass** (for smooth belts, C^0 convergence of embeddings with uniformly bounded area suffices), and let $\mathcal{F}_n \to \mathcal{F}$ **uniformly** on a neighborhood of the supports.

Proposition D.5 (stability of holonomy and phase).

Under the above hypotheses,

$$\|\mathcal{H}_n[S_n] - \mathcal{H}[S]\| \longrightarrow 0, \qquad W[S_n] \to W[S], \quad \text{and} \quad \Phi[S_n] \to \Phi[S] \text{ whenever } W[S] \neq 0.$$

Sketch. Apply Grönwall-type bounds to the Volterra series for $\mathcal{H}_n$ and use the uniform convergence $\mathcal{F}_n \to \mathcal{F}$ plus convergence of surface measures. The phase continuity follows from $W \neq 0$ and $\arg z$ being continuous off the branch cut. ■

D.5 Thin-belt limit and framing as a regular limit

Let $S_\varepsilon(\gamma, \nu)$ be the geodesic-normal belts of §D.1. Assume $\mathcal{F}$ is continuous near γ.

Proposition D.6 (existence of the thin-belt limit).

The limit $\lim_{\varepsilon \downarrow 0} \arg \mathrm{Tr}\,\mathcal{H}[S_\varepsilon(\gamma, \nu)]$ exists and depends only on the **framed loop** (γ, ν). Different framings ν, ν' can yield different limits (self-link/framing term).

Sketch. First-order area contributions are $O(\varepsilon)$, but the ordered-exponential boundary contraction in Appendix A produces a finite framing term controlled by the twist of ν (cf. Theorem A's minimality discussion). The limit is stable by Proposition D.5. ■

D.6 Operational measurability (pointer algebra & SBS)

With $W[S]$ and Φ Borel in γ and continuous under admissible limits, the **effects** "$\Phi \in I$" and "$\mathrm{Lk}_{\mathrm{belt}} = k$" are implementable as POVM outcomes written into the memory algebra $\mathcal{F}_{t+1}$. Under **commuting effects** and **accessible records** (SBS redundancy), SROQD's theorems give **AB-fixedness** and frame robustness (Collapse-Lorentz). We import these results to guarantee the **operational** status of our belt observables.

D.7 Summary (what this appendix guarantees)

Existence: Every admissible loop admits smooth **belts** $S_\varepsilon(\gamma, v)$ (Theorem D.1); for Lipschitz loops, **rectifiable** belts exist and can be smoothed (Theorem D.2).

Well-posedness: $\mathcal{H}[S]$ is defined on rectifiable belts by smooth approximation; $W[S]$ and Φ are Borel and stable under limits (Lemma D.3, Prop. D.5).

Thin-belt limit: Φ has a well-defined $\varepsilon \downarrow 0$ limit depending on the **framing** (Prop. D.6), matching the **minimality** statement of Theorem A in the draft.

These facts justify all regularity and convergence steps used in the main proofs (Theorems A–C) and in Appendix A's variation calculus, without any assumptions beyond the standard SK/Uhlmann framework already fixed in the paper.

Appendix E — Worked Reductions

We collect two explicit reductions that make the main theorems concrete: (i) the **Abelian/pure-state** limit where the belt calculus reproduces **Berry curvature forces**; (ii) 2+1**D Chern–Simons** examples where exchange phases become **framed Wilson** phases built from **explicit linking integrals**.

E.1 Abelian / pure-state reduction → Berry curvature forces

E.1.1 Setup and reduction

Let the transported state be rank-1, $P(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|$, over the parameter manifold Θ. The Uhlmann connection reduces to the Berry connection

$$\mathcal{A} \;\longrightarrow\; A := i\langle\psi \,|\, d\psi\rangle, \qquad \mathcal{F} \;\longrightarrow\; F := dA\,.$$

For a belt S spanning $\gamma_{\pm}$, the surface ordering drops out (Abelian case), so

$$\mathcal{H}[S] = \exp\Big(\iint_S F\Big), \qquad \Phi = \arg\operatorname{Tr}\,\mathcal{H}[S] = \iint_S F \pmod{2\pi}.$$

By Appendix A (Abelian form),

$$\delta\Phi = \oint_{\partial S} \iota_{\delta x} F, \qquad \Rightarrow \qquad F_i(\tau) \;=\; -\,\frac{\delta\,\Phi}{\delta x^i(\tau)} \;=\; -\,F_{ij}\big(x(\tau)\big)\,\dot{x}^j(\tau),$$

i.e. the **geometric force density** equals the Berry curvature contracted with the loop tangent.

E.1.2 Vector form (magnetic-analogue)

In local coordinates on $\Theta \subset \mathbb{R}^3$, set $B_k := \frac{1}{2}\epsilon_{kij}F_{ij}$. Then

$$\boxed{\mathbf{F}_{\text{geom}}(\tau) \;=\; -\,\dot{\mathbf{x}}(\tau)\times\mathbf{B}\big(\mathbf{x}(\tau)\big)\,.}$$

This is the familiar "Lorentz-like" anomalous response.

E.1.3 Two-level example (Bloch sphere monopole)

Let $H(\mathbf{R}) = \mathbf{R} \cdot \boldsymbol{\sigma}$ with $|\mathbf{R}| = R_0$; the ground band has Berry curvature

$$\mathbf{B}(\mathbf{R}) \;=\; -\,\frac{1}{2}\,\frac{\mathbf{R}}{|\mathbf{R}|^3}.$$

Hence

$$\mathbf{F}_{\mathrm{geom}}(\tau) = \;-\;\dot{\mathbf{R}}(\tau)\times\mathbf{B}(\mathbf{R}(\tau)) = \frac{1}{2}\,\dot{\mathbf{R}}(\tau)\times\frac{\mathbf{R}(\tau)}{|\mathbf{R}(\tau)|^3},$$

and the accumulated phase equals half the solid angle subtended by the loop on the Bloch sphere:

$$\Phi = \iint_S F = \;-\frac{1}{2}\,\Omega_{\mathrm{solid}}(S).$$

Everything here is a direct specialization of Theorem B to the Abelian/pure-state case.

E.2 2+1D Chern–Simons: framed Wilsons and explicit linking

We now evaluate the exchange phases in an Abelian Chern–Simons sector (the non-Abelian/Berry–Chern case follows by representation data). Level $k \in \mathbb{R} \setminus \{0\}$, action $S_{\mathrm{CS}}[A] = \frac{k}{4\pi}\int A \wedge dA$.

E.2.1 Gaussian evaluation with Poincaré-dual currents

For an oriented loop $C \subset \mathbb{R}^3$, let J_C be its Poincaré-dual 2-form: $\int \eta \wedge J_C = \oint_C \eta$. For (pairwise disjoint) loops C_a with charges q_a,

$$\left\langle \prod_a e^{\,iq_a \oint_{C_a} A} \right\rangle = \exp\left\{ i\,\frac{2\pi}{k}\Big(\sum_{a<b} q_a q_b\,\mathrm{Lk}(C_a, C_b) + \frac{1}{2}\sum_a q_a^2\,\mathrm{SL}(C_a)\Big)\right\},$$

where:

$\mathrm{Lk}(C_1, C_2)$ is the Gauss **linking number**,

$$\mathrm{Lk}(C_1, C_2) = \frac{1}{4\pi}\oint_{C_1}\oint_{C_2} \frac{(\mathbf{r}_1 - \mathbf{r}_2)\cdot(d\,\mathbf{r}_1 \times d\,\mathbf{r}_2)}{\|\mathbf{r}_1 - \mathbf{r}_2\|^3},$$

$\mathrm{SL}(C) \in \mathbb{Z}$ is the **self-link** w.r.t. a chosen ribbon framing (Călugăreanu–White: SL = Wr + Tw).

Belts. Each excitation is represented by an SK **belt** with two framed boundary components $\gamma_\pm$. Summing over all boundary pairs gives the **framed belt linking**

$$\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) = \sum_{\alpha,\beta\,\in\,\{+,-\}} \mathrm{Lk}(\gamma_{1,\alpha}, \gamma_{2,\beta}).$$

Since each $\gamma_{j,\pm}$ is a small framed offset of the core loop C_j, $\mathrm{Lk}(\gamma_{1,\alpha}, \gamma_{2,\beta}) = \mathrm{Lk}(C_1, C_2)$ for all α, β, hence

$$\boxed{\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) = 4\,\mathrm{Lk}(C_1, C_2)}$$

for consistent SK orientations. The overall coefficient is absorbed into the sector constant κ of the main text.

Thus the **exchange unitary** between two belts is

$$U_{\mathrm{ex}} = \exp\left(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2)\right)\cdot U_{\mathrm{local}}, \qquad \kappa = \frac{2\pi}{k}\, q_1 q_2 \ \text{(Abelian case)}.$$

Self-link terms $\frac{1}{2}\sum_{j,\alpha} q_{j,\alpha}^2\,\mathrm{SL}(\gamma_{j,\alpha})$ produce the **framing anomaly** (a calibration offset fixed by the SK ribbon framing).

E.2.2 Explicit Hopf-link example (linked once)

Take two core circles:

$$C_1\colon\ (\cos u,\ \sin u,\ 0), \quad C_2\colon\ (2,\ \cos v,\ \sin v), \qquad u, v \in [0, 2\pi).$$

They form a Hopf link with $\mathrm{Lk}(C_1, C_2) = 1$. Offsetting by a small normal field yields the belt boundaries $\gamma_{j,\pm}$. Therefore

$$\mathrm{Lk}_{\mathrm{belt}}(S_1, S_2) = 4, \qquad U_{\mathrm{ex}} = \exp\left(i\,4\,\kappa\right)\cdot U_{\mathrm{local}}\,.$$

If $q_1 = q_2 = 1$ and level k, then $\kappa = 2\pi\,/\,k$ and

$$U_{\mathrm{ex}} = \exp\left(i\,\frac{8\pi}{k}\right)\cdot U_{\mathrm{local}}\,.$$

E.2.3 Unlinked loops and framing calibration

If C_1, C_2 are unlinked, $\mathrm{Lk}(C_1, C_2) = 0$ so $\mathrm{Lk}_{\mathrm{belt}} = 0$ and only the **self-link** phases remain:

$$\langle W[\partial S_1]\, W[\partial S_2]\rangle = \exp\left\{i\,\frac{2\pi}{k}\ \frac{1}{2}\sum_{j,\alpha} q_{j,\alpha}^2\,\mathrm{SL}(\gamma_{j,\alpha})\right\}.$$

Changing the belt framing by one twist unit shifts $\mathrm{SL} \mapsto \mathrm{SL} \pm 1$, adding a constant phase; mutual phases are unaffected. This is the standard framing anomaly and is fixed once the SK belt framing is chosen.

E.2.4 Non-Abelian/Berry–Chern remark

For non-Abelian sectors or adiabatic bands with Berry–Chern response, replace charges q_a by representation data (e.g., quadratic Casimir factors) and k by the appropriate **level/normalization** of the topological term. The framed-linking structure and the belt summation remain identical; the sector constant κ encodes those details.

Summary.

In the **Abelian/pure-state** limit, Theorem B delivers the usual **Berry curvature force** $F_i = -F_{ij}\dot{x}^j$.

In 2+1**D Chern–Simons** sectors, Theorem C yields explicit **framed Wilson** phases: exchange $= \exp(i\,\kappa\,\mathrm{Lk}_{\mathrm{belt}})$ with self-link (framing anomaly) fixed by the SK ribbon. These calculations exemplify the paper's central equivalences in concrete, computable settings.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.